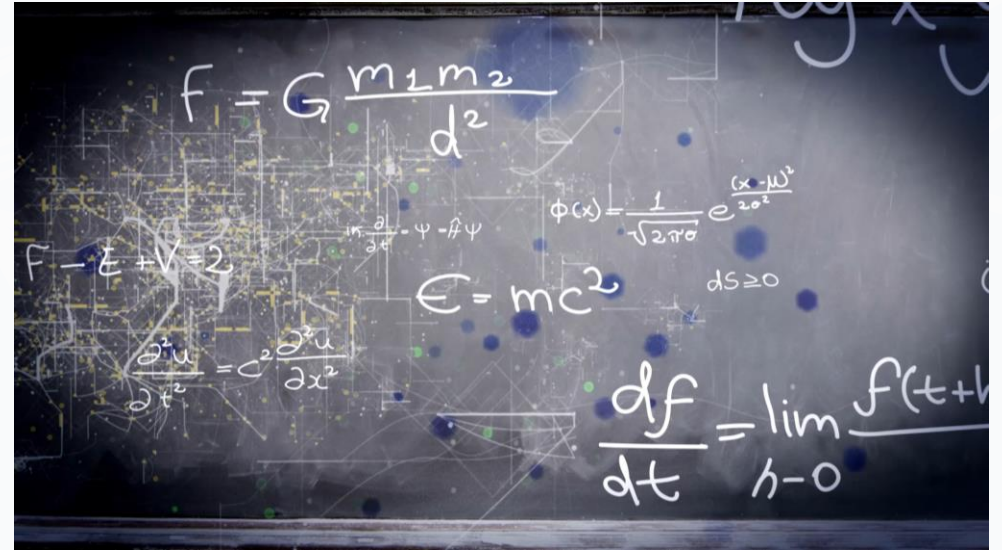


Machine Learning

Classical algorithms –
Overview



GROKKERS
AI FOR EVERYONE

Topics

Introduction to Machine Learning

DATA for Machine Learning

What is a MODEL?

Types of Machine Learning

Typical tasks from a ML model

ML Platforms

Pre-reqs



Python basics (language)



NumPy, PANDAS, sciPy, matplotlib, seaborn etc



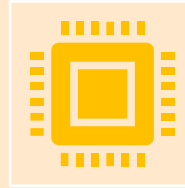
Statistics essentials for Data Science



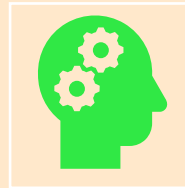
Must have gone thru data pre-processing steps and concepts on data splits

Definition

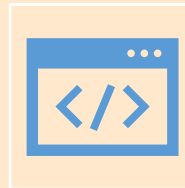
(let us learn from examples)



Machine learning is a part of AI technology that enables computers to learn and understand patterns from data



Use the learning to make predictions or decisions



improve its performance over time without explicit programming.

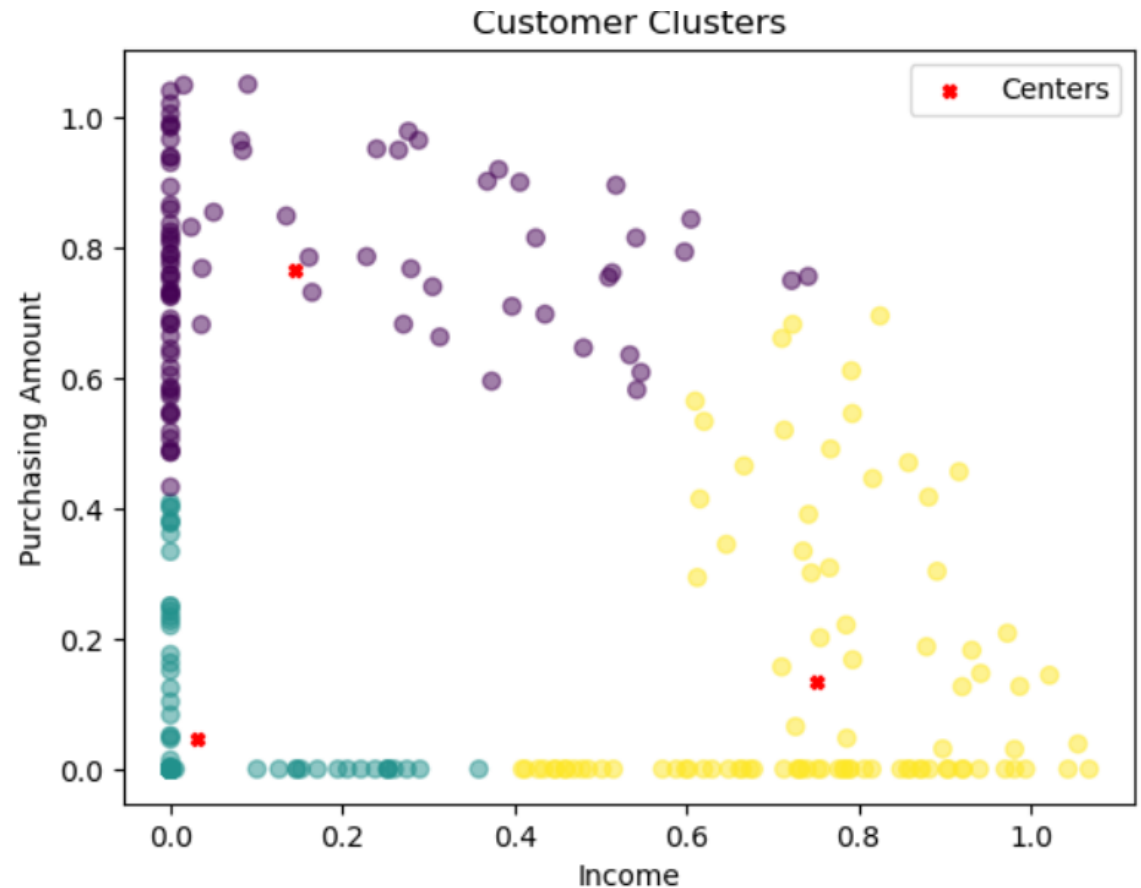
Examples



Customer Segmentation:

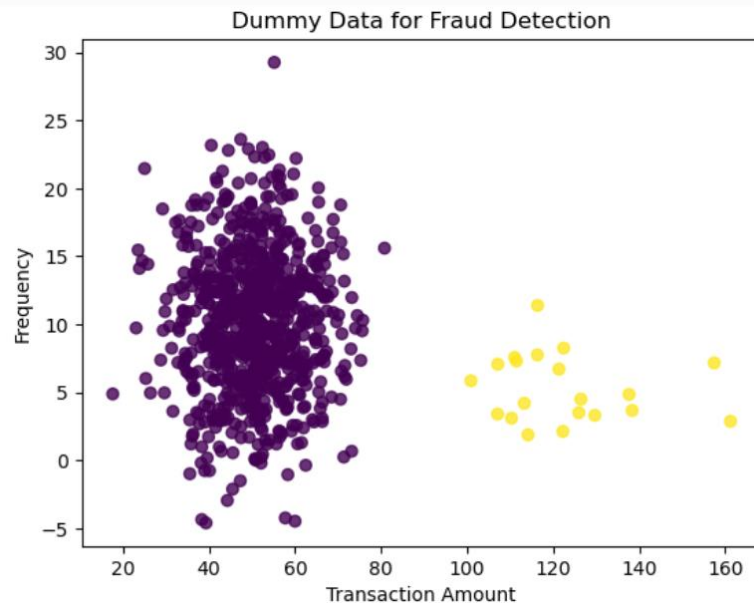
Scenario: A retail business wants to understand its customer base better.

Clustering algorithms can analyze customer data to identify distinct segments based on purchasing behavior, demographics, or preferences.



Fraud Detection

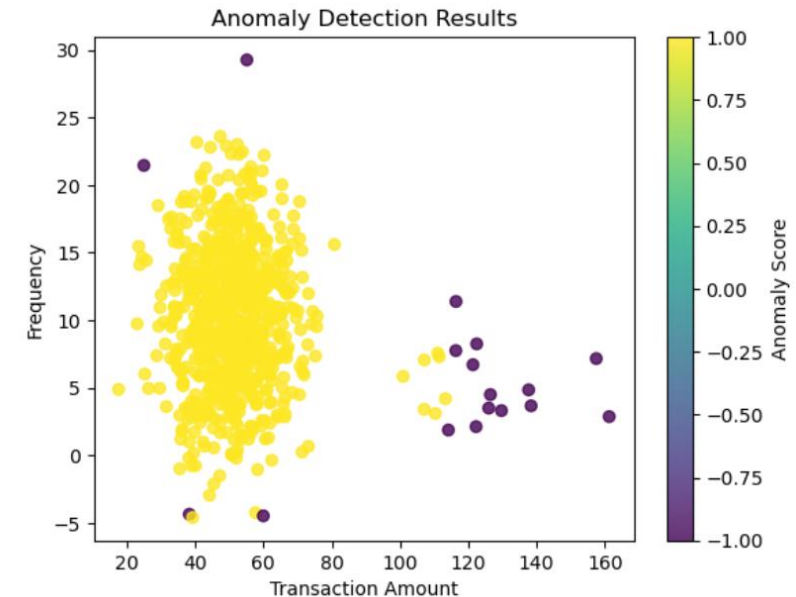
- **Scenario:** A financial institution aims to detect fraudulent transactions.
- Anomaly detection algorithms can learn from historical transaction data to identify patterns indicative of fraudulent activity, flagging suspicious transactions for further review.



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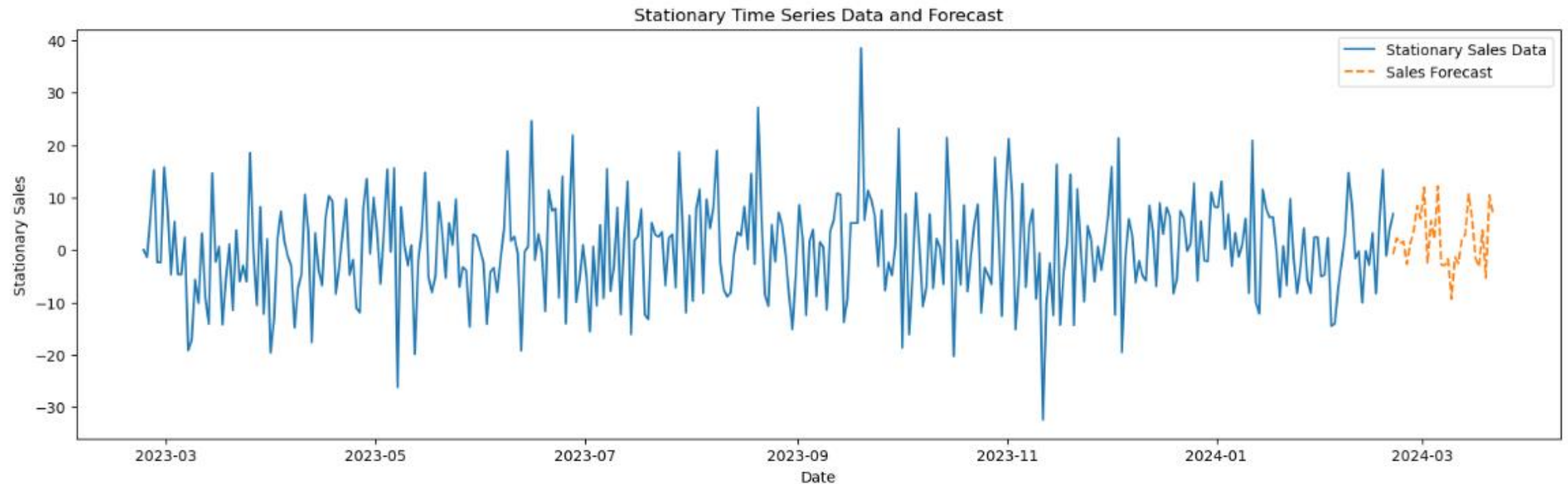


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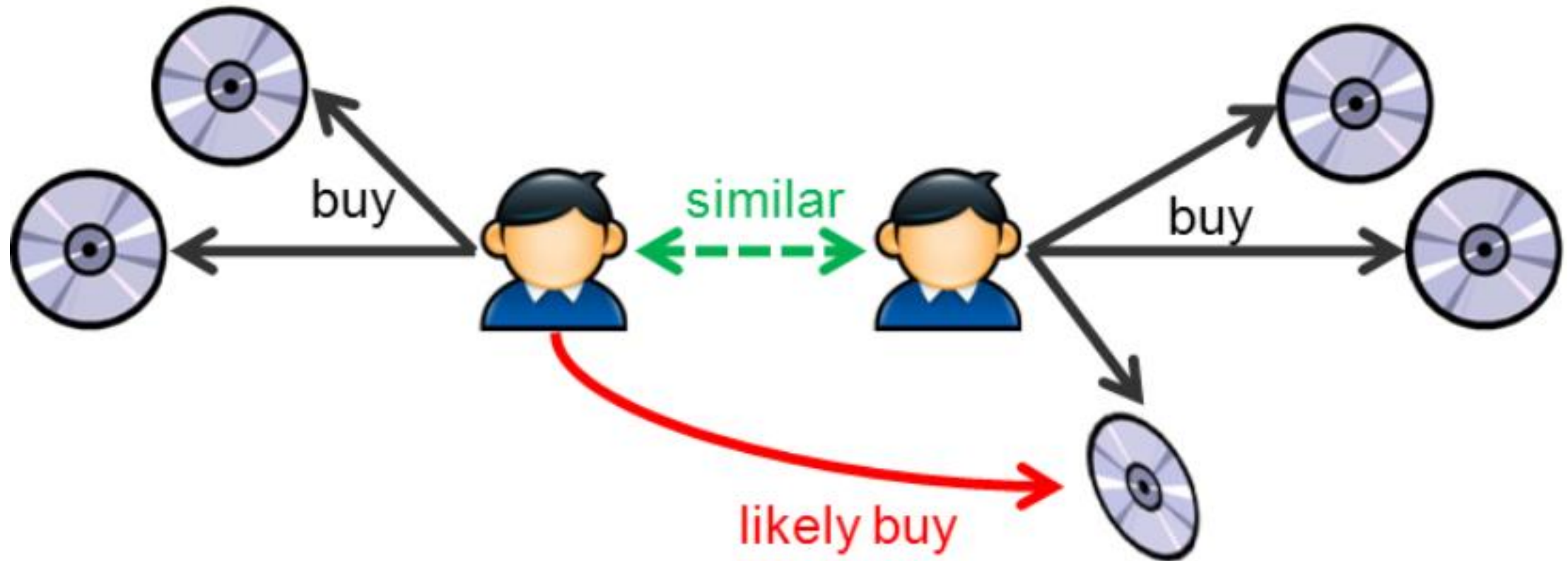
Demand Forecasting

- **Scenario:** An e-commerce company needs to optimize inventory management.
- Time series forecasting models can analyze historical sales data to predict future demand, allowing the business to stock the right amount of inventory.



Personalized Marketing

- **Scenario:** A marketing team wants to improve campaign effectiveness.
- Recommender systems can analyze customer preferences and behavior to recommend personalized products or services, increasing the likelihood of conversion.



Text Classification



Sentiment Analysis:

Determine whether a given text expresses a positive, negative, or neutral sentiment.



Topic Classification: Categorize articles or documents into predefined topics.



Spam Detection: Identify spam emails or messages.

Examples of sentiment analysis



Product Reviews:

Text: "I love my new smartphone! The camera quality is amazing."

Sentiment: Positive

Text: "The laptop's battery life is disappointing. I regret buying it."

Sentiment: Negative



Social Media Posts:

Text: "Just booked my dream vacation! So excited!"

Sentiment: Positive

Text: "Feeling frustrated with the terrible customer service from XYZ company."

Sentiment: Negative



Movie Reviews:

Text: "This movie is a masterpiece. The acting and storyline are exceptional."

Sentiment: Positive

Text: "The film was a complete letdown. Poorly written and badly directed."

Sentiment: Negative

Text Summarization



Extractive Summarization: Extract salient sentences or phrases from a text to create a summary.



Abstractive Summarization: Generate a concise and coherent summary using new sentences.

Machine Translation

- is a sub-field of computational linguistics that focuses on translating text or speech from one language to another using algorithms, without human intervention.
- MT can be rule-based, statistical, neural, or a hybrid of these techniques.
- **Input (English):** "Hello, how are you?"
- Using a machine translation system (like Google Translate, DeepL, or a custom-trained model), you get:
- **Output (French):** "Bonjour, comment ça va?"



Named Entity Recognition (NER)

is a sub-task of information extraction in Natural Language Processing (NLP) that classifies named entities into predefined categories such as names of persons,

Sentence: "Barack Obama was born in Hawaii."

1. "Barack Obama" → **PERSON**
2. "Hawaii" → **LOCATION**

Sentence: "Apple Inc. is planning to open a new store in San Francisco by January 2023."

1. "Apple Inc." → **ORGANIZATION**
2. "San Francisco" → **LOCATION**
3. "January 2023" → **TIME**

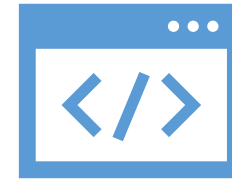
Question Answering



focuses on building systems that can automatically answer questions posed by humans in a natural language format.



In essence, QA systems can understand and retrieve precise answers from a given text (or a collection of texts) based on a posed question.



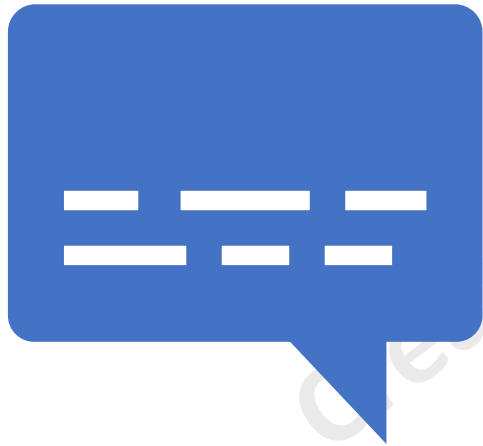
This is different from traditional search engines that return a list of documents or snippets - QA aims to return exact answers.



Example

- **Given Text (Passage):** "Mount Everest is the Earth's highest mountain, with a peak at 8,848.86 meters above sea level. It is located in the Mahalangur Himal sub-range of the Himalayas on the border between China and Nepal. Sir Edmund Hillary from New Zealand and Tenzing Norgay, a Sherpa of Nepal, were the first to reach the summit on 29 May 1953."
- **Question:** "Who were the first people to reach the summit of Mount Everest?"
- **QA System Output:** "Sir Edmund Hillary from New Zealand and Tenzing Norgay, a Sherpa of Nepal, were the first to reach the summit on 29 May 1953."

Text Generation



- refers to the process of automatically producing human-readable and coherent text that is contextually relevant, often based on some input.
- This task is a significant area of research within Natural Language Processing (NLP) and has been bolstered with the advent of deep learning models, particularly recurrent neural networks (RNNs), long short-term memory networks (LSTMs), and more recently, Transformer-based models like GPT-2 and GPT-3.

Example

- **Prompt:** "Once upon a time, in a land far away,"
- **Possible Text Generation Output:** "Once upon a time, in a land far away, there was a small village nestled between two great mountains. The villagers led a simple life, farming the fertile plains and trading with nearby towns. However, a legend spoke of a hidden treasure in the heart of the mountain, guarded by a fierce dragon. Many adventurers tried to find this treasure but none returned. It was only when a young maiden named Lila, with courage in her heart and a song on her lips, ventured into the mountain that the fate of the village began to change."

Another useful application

Data Augmentation:

Generate variations of existing text data to enhance datasets, especially useful in training machine learning models.



Information Retrieval

- **Search Engines:** Retrieve relevant documents or web pages based on user queries.
- **Document Similarity:** Identify similar documents in a large corpus.

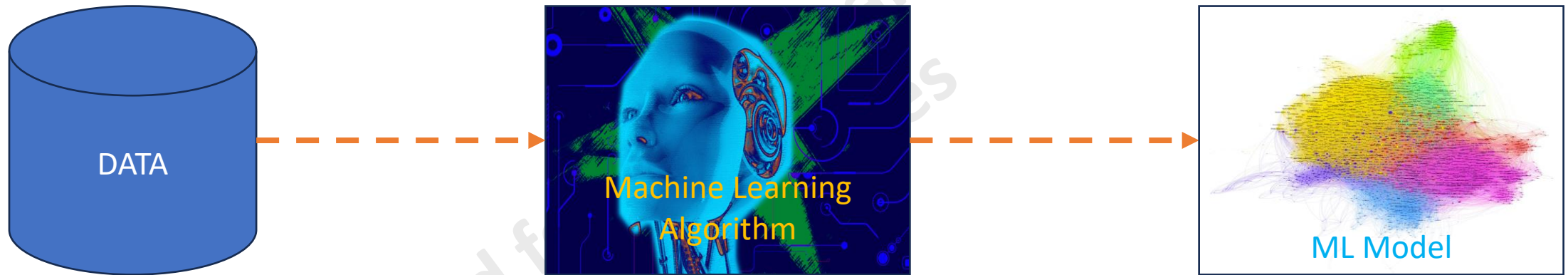


Other important NLP applications

1. **Keyword/Keyphrase Extraction:** Extract important terms or phrases from a text.
- **Relation Extraction:** Identify and classify relationships between named entities.
 - **Part-of-Speech Tagging:** Label words in a sentence with their appropriate part-of-speech markers (like noun, verb, adjective, etc.).



ML needs ... Data ... to produce a model



- Neighbors' algorithm
- Naïve Bayes
- Decision Trees
- ...
- Neural Nets

What does the DATA look like ?



For Machine Learning

Sample dataset (temp forecasts)

year	month	day	week	Temp 2 days before	Temp 1 day before	Average temp	Actual temp on that day	Temp forecast by noaa	Temp Forecast by acc	Temp forecast by friend
2016	1	1	Fri	45	45	45.6	45	43	50	29
2016	1	2	Sat	44	45	45.7	44	41	50	61
2016	1	3	Sun	45	44	45.8	41	43	46	56
2016	1	4	Mon	44	41	45.9	40	44	48	53
2016	1	5	Tues	41	40	46	44	46	46	41
2016	1	6	Wed	40	44	46.1	51	43	49	40
2016	1	7	Thurs	44	51	46.2	45	45	49	38
2016	1	8	Fri	51	45	46.3	48	43	47	34
2016	1	9	Sat	45	48	46.4	50	46	50	47
...	...	...	...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...	...	...
2016	1	19	Tues	50	54	47.6	48	47	49	53
2016	1	20	Wed	54	48	47.7	52	44	52	61
2016	1	21	Thurs	48	52	47.8	52	43	51	57

Independent variables

23
Independent variables

Dependent variable

Sample dataset (loan applications)

Loan_ID	Gender	Married	Number of Dependents	Education	Self Employed ?	Applicant Income	Co-applicant Income	Loan Amount	Loan Amount Term	Credit History	Property Area
LP001032	Male	No	0	Graduate	No	4950	0	125	360	1	Urban
LP001824	Male	Yes	1	Graduate	No	2882	1843	123	480	1	Semiurban
LP002928	Male	Yes	0	Graduate	No	3000	3416	56	180	1	Semiurban
LP001814	Male	Yes	2	Graduate	No	9703	0	112	360	1	Urban
LP002244	Male	Yes	0	Graduate	No	2333	2417	136	360	1	Urban
LP001854	Male	Yes	3+	Graduate	No	5250	0	94	360	1	Urban
...	Male	Yes	0	Graduate	No	3500	1667	114	360	1	Semiurban
LP001647	Male	Yes	0	Graduate	No	9328	0	188	180	1	Rural
LP001871	Female	No	0	Graduate	No	7200	0	120	360	1	Rural
LP001379	Male	Yes	2	Graduate	No	3800	3600	216	360	0	Urban
LP002789	Male	Yes	0	Graduate	No	3593	4266	132	180	0	Rural
LP001578	Male	Yes	0	Graduate	No	2439	3333	129	360	1	Rural
LP001318	Male	Yes	2	Graduate	No	6250	5654	188	180	1	Semiurban
LP001259	Male	Yes	1	Graduate	Yes	1000	3022	110	360	1	Urban
LP002804	Female	Yes	0	Graduate	No	4180	2306	182	360	1	Semiurban

Independent variables

24

Independent variables

Dependent variable ?

SAMPLE DATASET (automobiles)

KMs per liter	cylinders	displacement	horsepower	weight	acceleration	year	origin	Model name
18	8	307	130	3504	12	70	1	chevrolet chevelle malibu
15	8	350	165	3693	11.5	70	1	buick skylark 320
18	8	318	150	3436	11	70	1	plymouth satellite
16	8	304	150	3433	12	70	1	amc rebel sst
17	8	302	140	3449	10.5	70	1	ford torino
15	8	429	198	4341	10	70	1	ford galaxie 500
...	...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...
24	4	113	95	2372	15	70	3	toyota corona mark ii
22	6	198	95	2833	15.5	70	1	plymouth duster
18	6	199	97	2774	15.5	70	1	amc hornet
21	6	200	85	2587	16	70	1	ford maverick
27	4	97	88	2130	14.5	70	3	datson pl510

Dependent variable ?

Sample dataset (diabetes related)

Pregnancies	Glucose	Blood Pressure	Skin Thickness	Insulin	BMI	Diabetes Pedigree Function	Age	Diabetic ? (1 = Yes, 0= No)
6	148	72	35	0	33.6	0.627	50	1
1	85	66	29	0	26.6	0.351	31	0
8	183	64	0	0	23.3	0.672	32	1
1	89	66	23	94	28.1	0.167	21	0
0	137	40	35	168	43.1	2.288	33	1
5	116	74	0	0	25.6	0.201	30	0
3	78	50	32	88	31	0.248	26	1
10	115	0	0	0	35.3	0.134	29	0
2	197	70	45	543	30.5	0.158	53	1
8	125	96	0	0	0	0.232	54	1
4	110	92	0	0	37.6	0.191	30	0
10	168	74	0	0	38	0.537	34	1
10	139	80	0	0	27.1	1.441	57	0

Dependent variable ?

example

year	month	day	week	Temp 2 days before	Temp 1 day before	Average temp	Actual temp on that day	Temp forecast by <u>noaa</u>	Temp Forecast by acc	Temp forecast by friend
2016	1	1	Fri	45	45	45.6	45	43	50	29
2016	1	2	Sat	44	45	45.7	44	41	50	61
2016	1	3	Sun	45	44	45.8	41	43	46	56
2016	1	4	Mon	44	41	45.9	40	44	48	53
2016	1	5	Tues	41	40	46	44	46	46	41
2016	1	6	Wed	40	44	46.1	51	43	49	40
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...	...	...	...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...	...	...
2016	1	19	Tues	50	54	47.6	48	47	49	53
2016	1	20	Wed	54	48	47.7	52	44	52	61
2016	1	21	Thurs	48	52	47.8	52	43	51	57

Independent variables

Independent variables

Dependent variable

What does the ML model look like

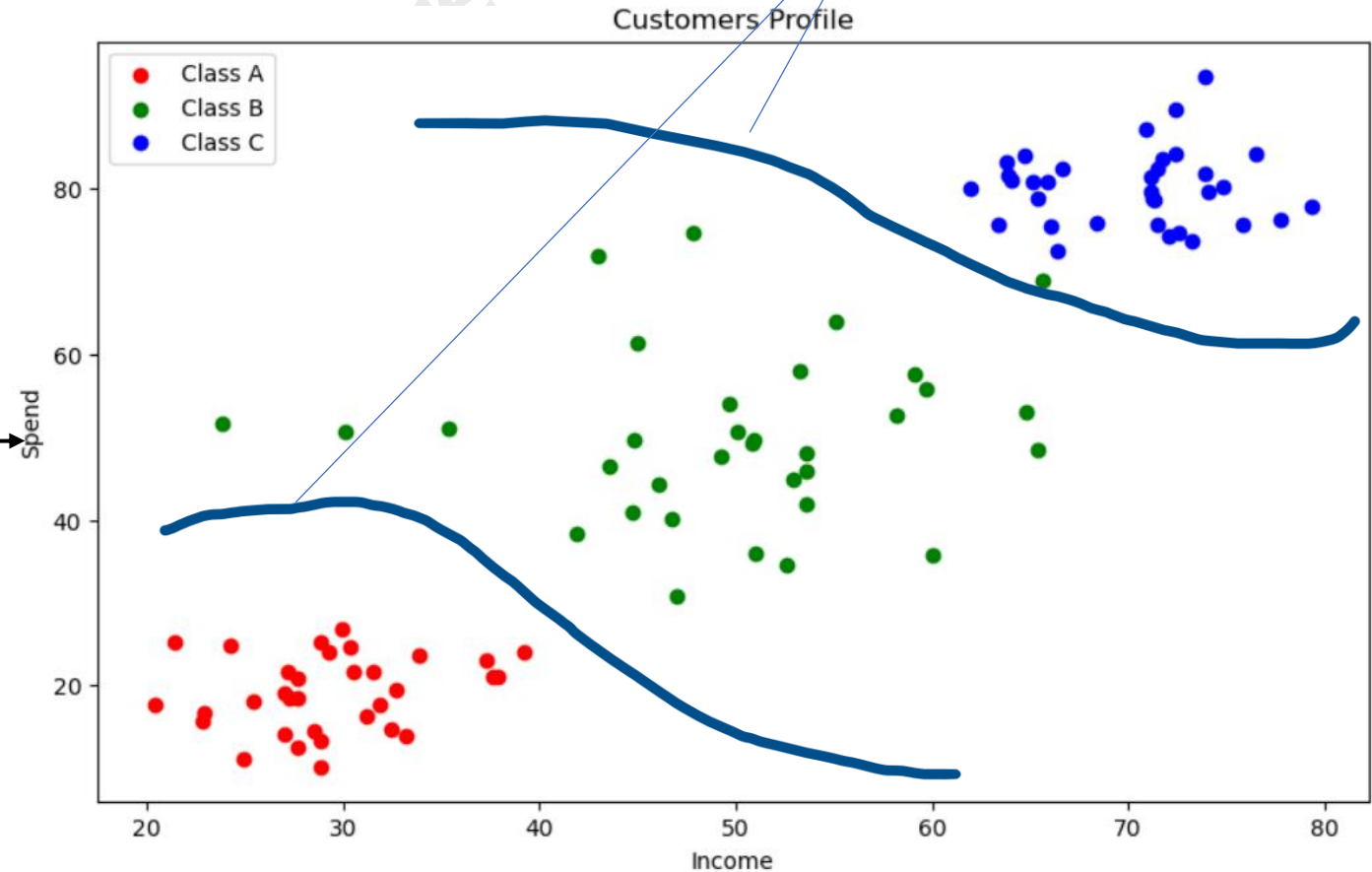
Intuitively

Customer segments

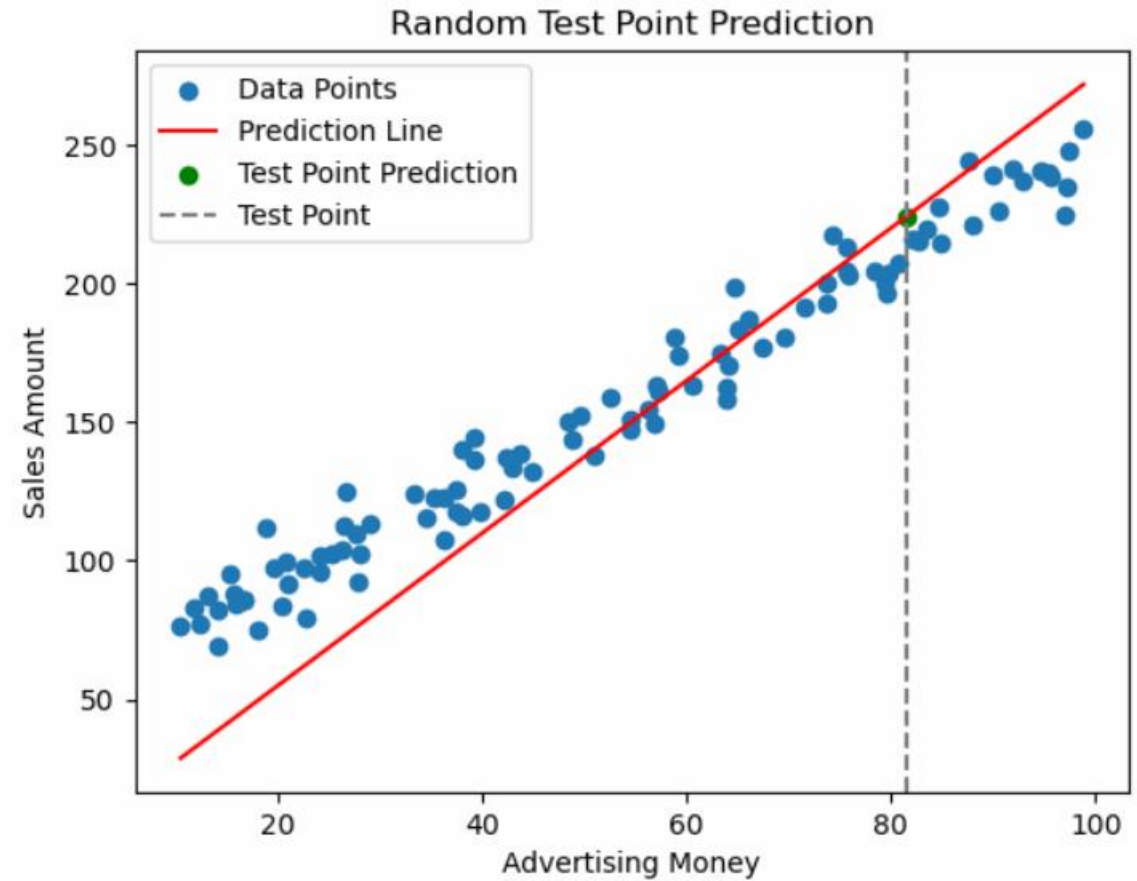
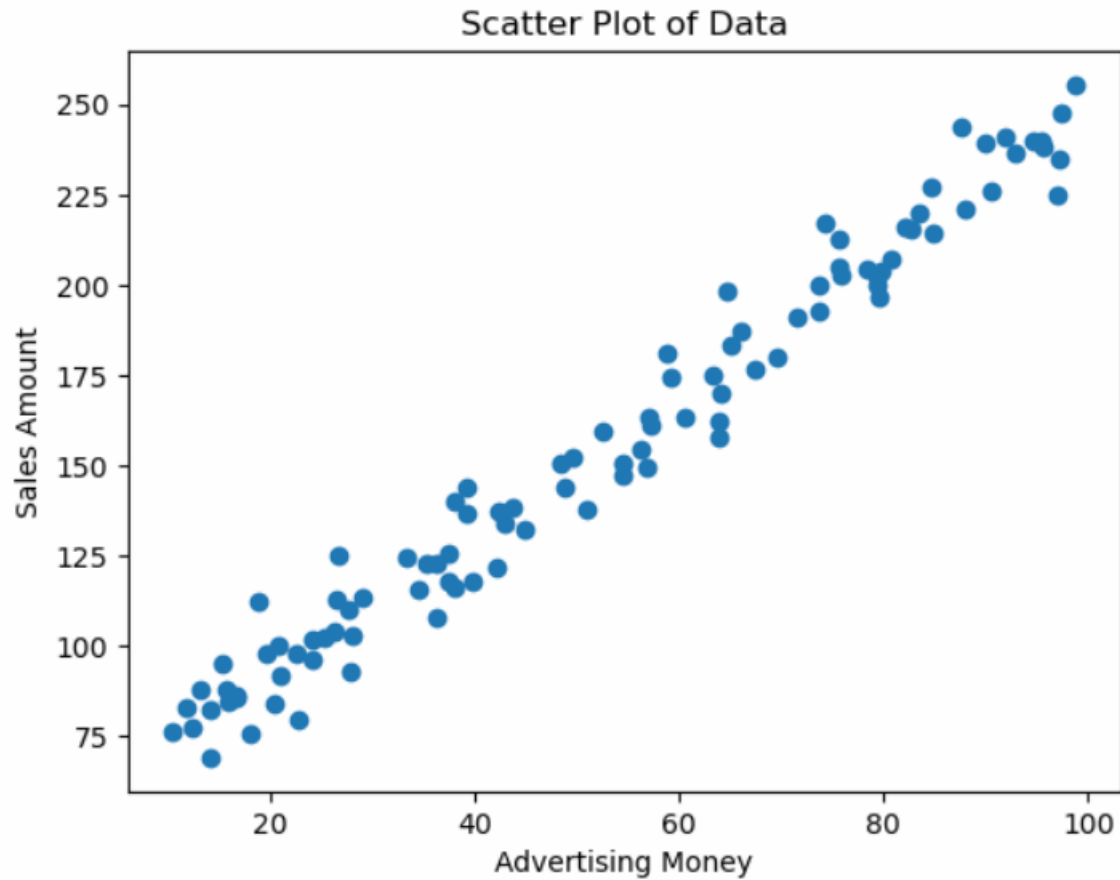
- ML Model (learns the pattern in Data)

	Income	Spend	Class
0	32.483571	14.711445	A
1	29.308678	24.112725	A
2	33.238443	13.895782	A
3	37.615149	21.044318	A
4	28.829233	10.201649	A
...	...	...	...
95	73.935423	81.926587	C
96	75.792978	75.580713	C
97	65.896588	80.768626	C
98	74.816881	80.291044	C
99	72.063905	74.285149	C

ML
algorithm



Sales prediction

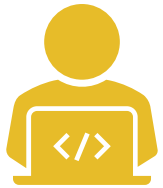


Define Machine learning

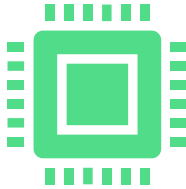


How it evolved, process etc, Simple example of ML flow

Traditional Programming



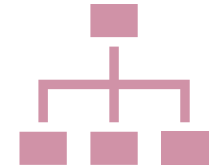
Rule-based Logic: explicit instructions (rules) for the computer to follow.



Deterministic: produce the same output for a given input every time.



Limited Adaptation: If the input or context changes, the program might **not** work correctly.



Specific Problem Solving: Traditional programs are designed for specific tasks and often require human intervention to handle changes or new situations.

Machine Learning

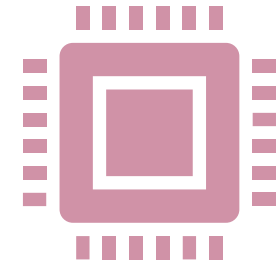


Learning from Data computers learn from data patterns instead of following explicit rules.



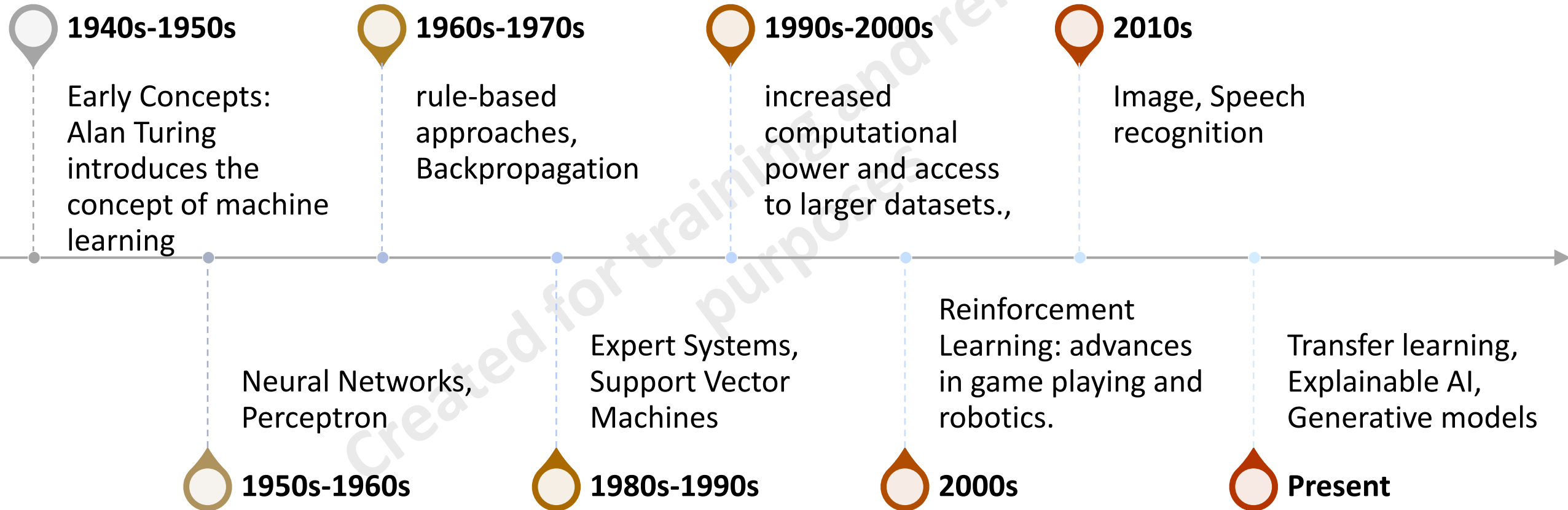
Probabilistic

make predictions based on probabilities. They provide confidence scores for their predictions.



Adaptation and Learning adapt and improve over time as they're exposed to more data.

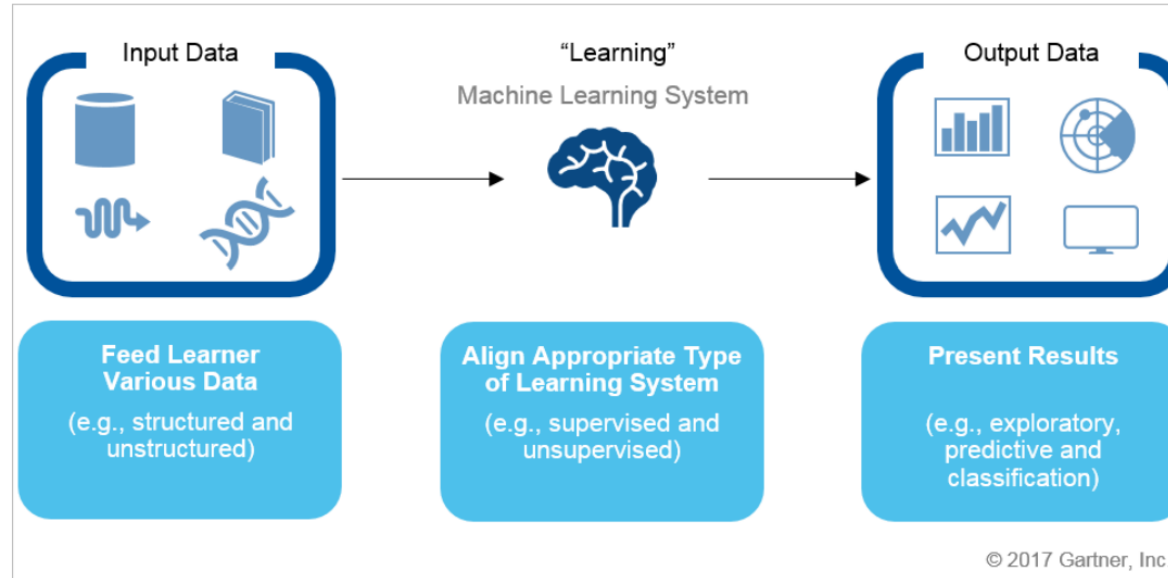
timeline



Machine Learning Process Overview

Input data: A wide variety of data can be used as input for ML purposes, variety of sources, enterprise, mainframes, IOT devices.

Formats – structured or unstructured.

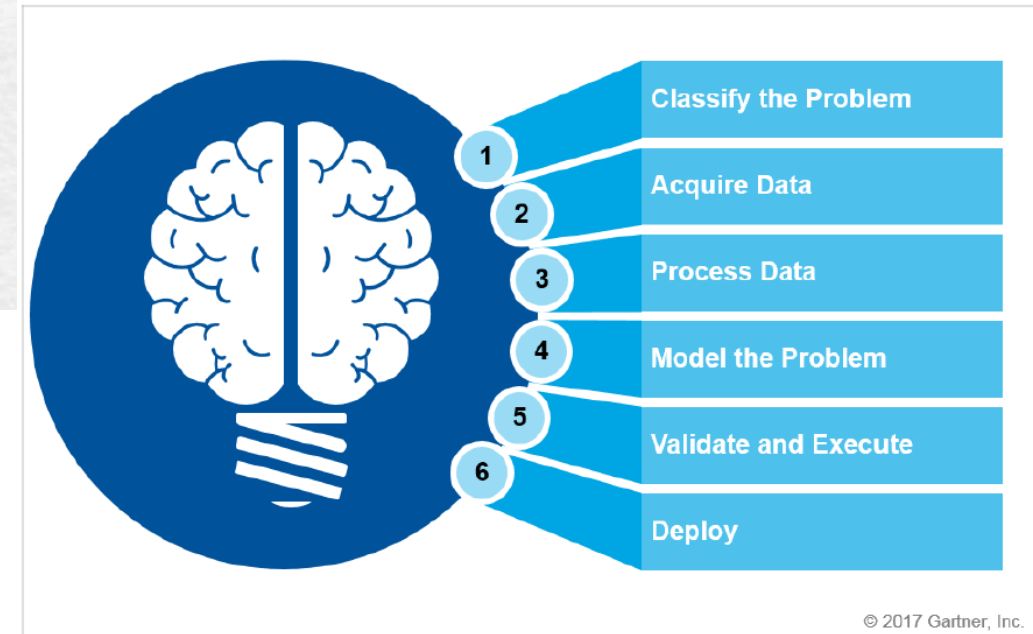
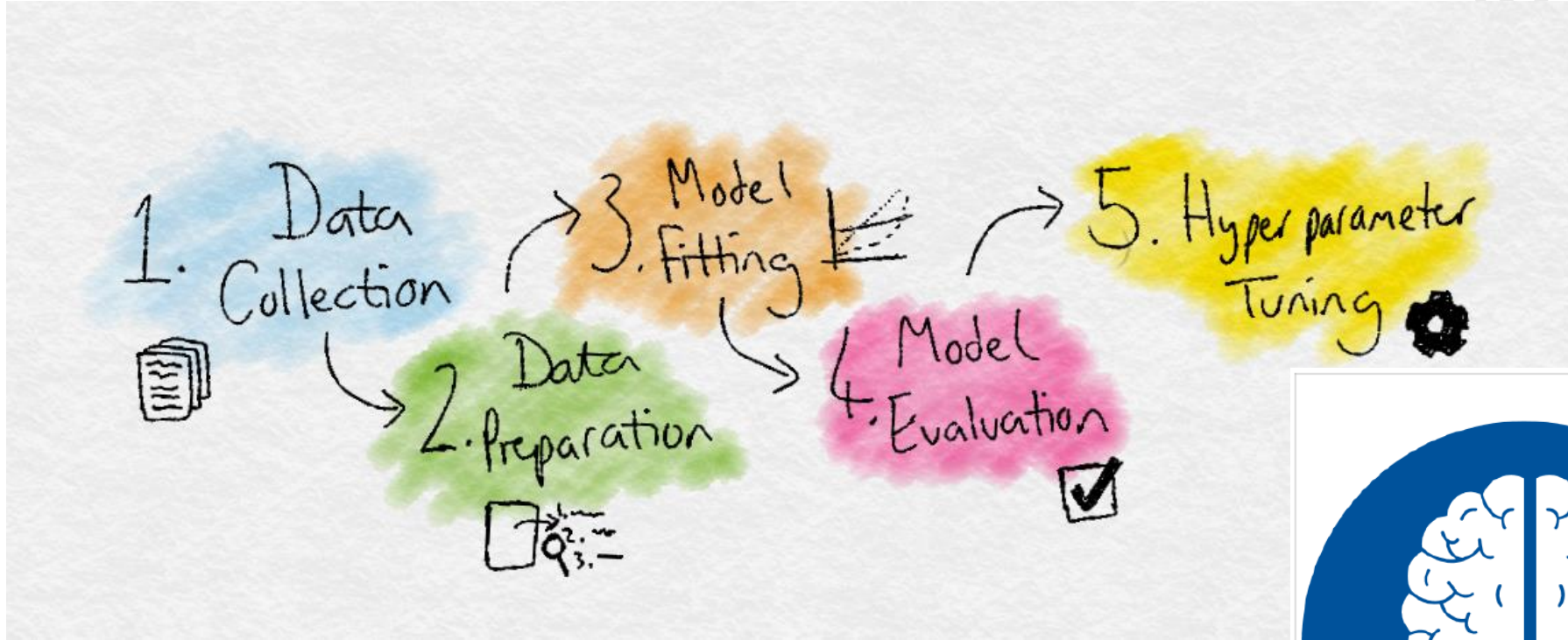


Source: Gartner (January 2017)

Output data: ML can be used to deliver results that are either predictive (that is, providing forecasts) or prescriptive (that is, suggesting recommended actions).

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Stages of ml



© 2017 Gartner, Inc.

The stages of ML often overlaps with the data science life cycle

Data science life cycle

Task (Proportion of Effort)	Subtasks	Stakeholder		
		Business	Data Scientist	IT/ Operations
1. Problem Understanding (5% to 10%)	a) Determine Objective	X	X	
	b) Define Success Criteria	X	X	
	c) Assess Constraints	X	X	X
2. Data Understanding (10% to 25%)	a) Assess Data Situation	X	X	X
	b) Obtain Data (Access)		X	X
	c) Explore Data	X	X	X
3. Data Preparation (20% to 40%)	a) Filter Data		X	X
	b) Clean Data		X	X
	c) Feature Engineering	X	X	
4. Modeling (20% to 30%)	a) Select Model Approach		X	
	b) Build Models		X	
5. Evaluation of Results (5% to 10%)	a) Select Model		X	
	b) Validate Model		X	
	c) Explain Model	X	X	
6. Deployment (5% to 15%)	a) Deploy Model		X	X
	b) Monitor and Maintain	X	X	X
	c) Terminate	X	X	X

Source: Gartner (January 2017)

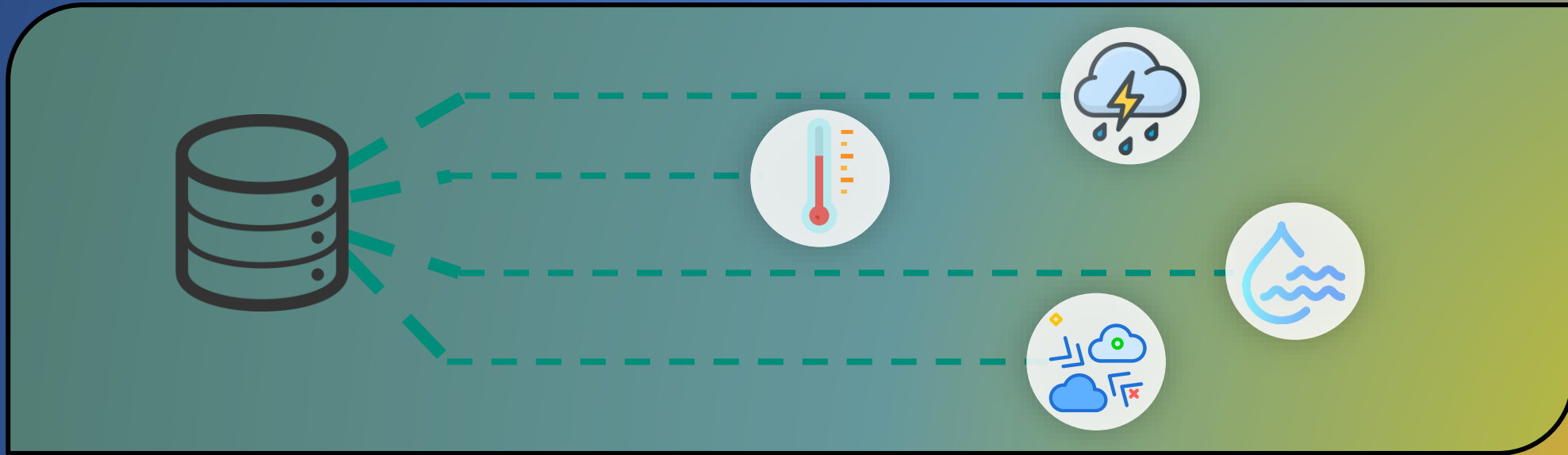
Step 1: Define the objective of the Problem

- What are we trying to predict?
- What are the target features?
- What is the input data?
- What kind of problem are we facing? Binary classification?
Clustering?



To predict the possibility of rain by studying the weather conditions.

Step 2: Data Gathering



manData such as weather conditions, humidity level, temperature, pressure, etc are either collected usually or scarped from the web.

Step 3: Preparing Data

- Transform data into desired format
 - Numeric, Text, Audio, Videos, PDFs, Docs, CSVs
 - Time Series, Graph Data
- Data cleaning
 - Missing values, anomalies
 - Corrupted data, Remove unnecessary data
 - Aggregation
 - Feature Engineering



Data Cleaning involves getting rid of inconsistencies in data such as missing values or redundant variables.

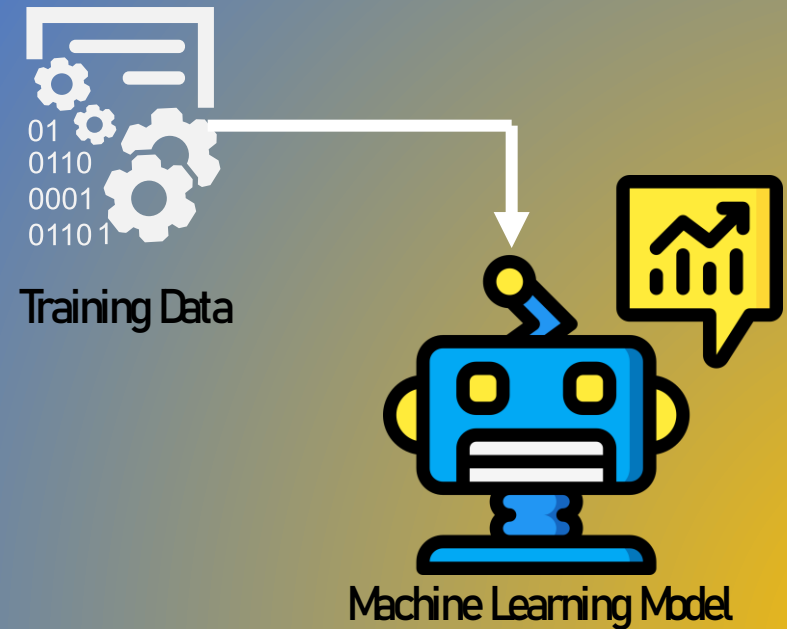
Step 4: Exploratory Data Analysis



Data Exploration involves understanding the patterns and trends in the data. At this stage all the useful insights are drawn and correlations between the variables are understood.

Step 5: Building a Machine Learning Model

- Machine Learning model is built by using the training data set
- The model is the Machine Learning algorithm that predicts by using the historical data fed to it



Predictive Model is built by using Machine Learning Algorithms such as Linear Regression, Decision Trees, etc.

Logical models

Tree models and Rule models.

Concept learning

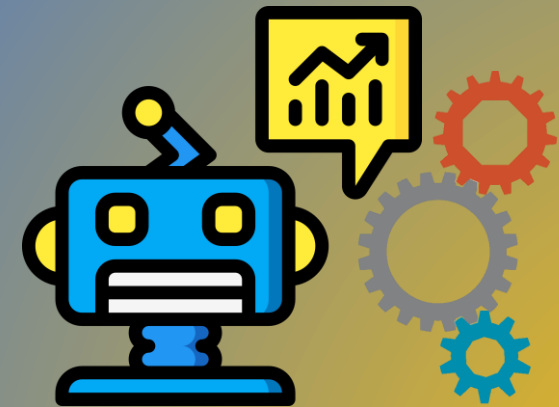
Geometric models

Linear models, Distance-based models

Probabilistic models

Step 6: Model Evaluation & Optimization

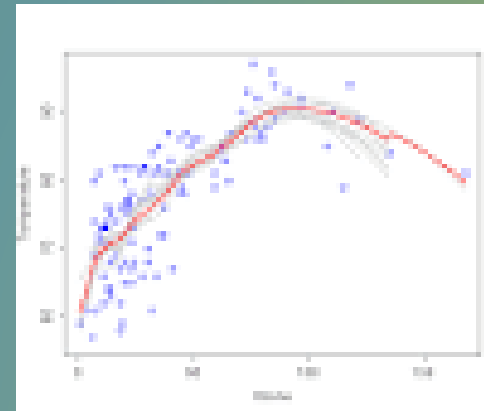
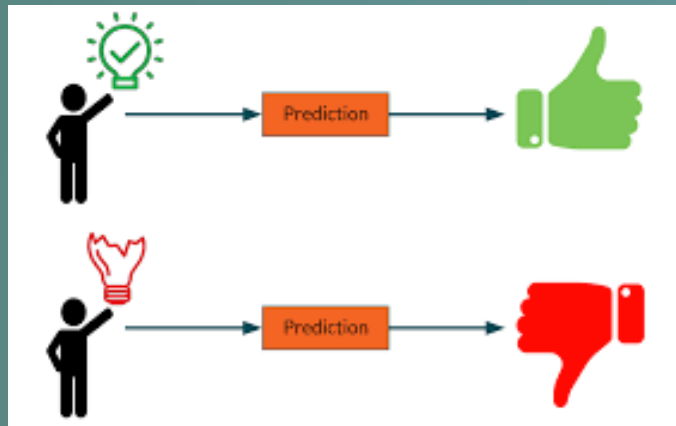
- Machine Learning model is evaluated by using the testing data set
- The accuracy of the model is calculated
- Further improvement in the model are done by using techniques like Parameter tuning



Machine Learning Model

The efficiency of the model is evaluated and any further improvement in the model are implemented.

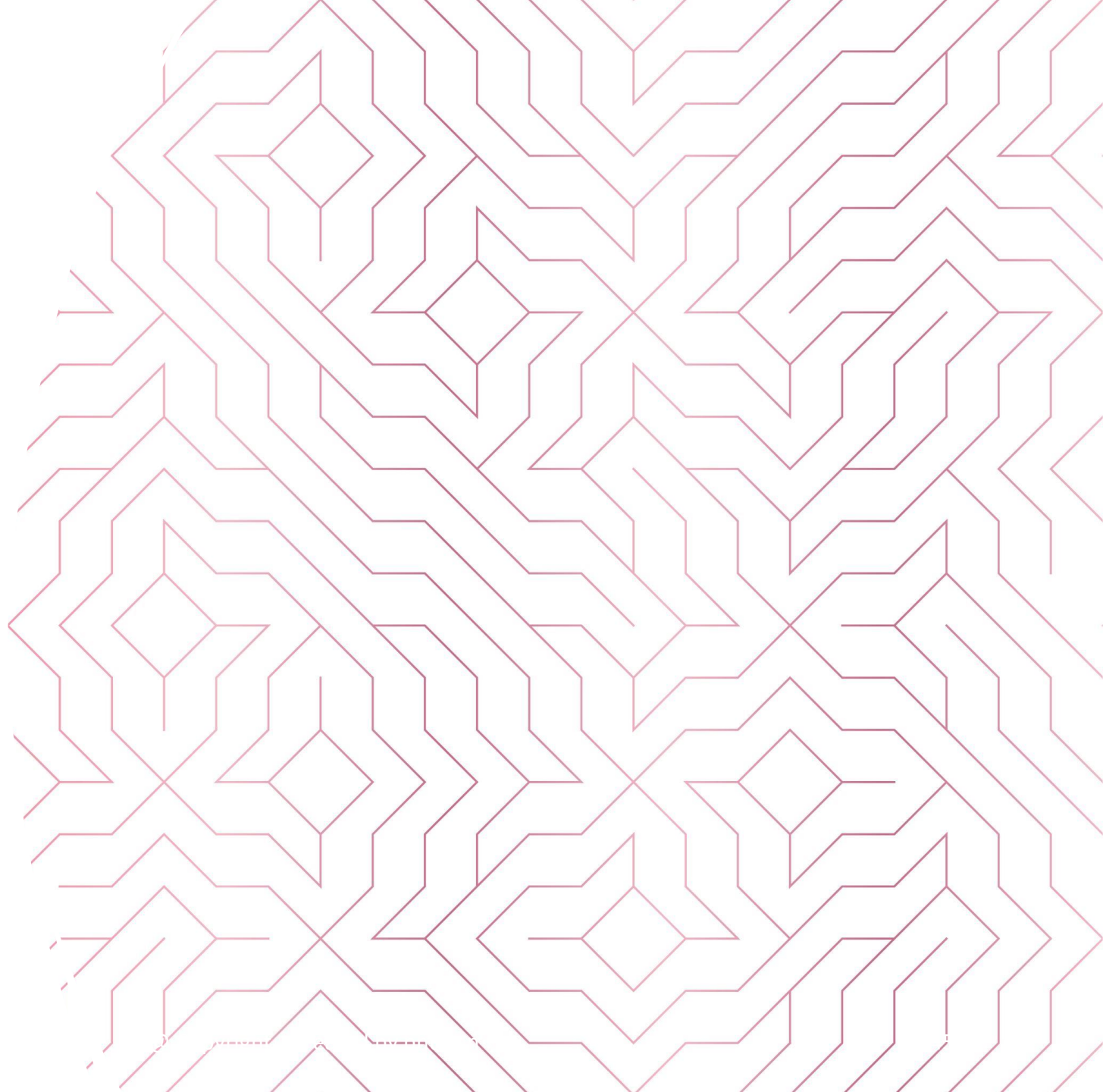
Step 7: Predictions



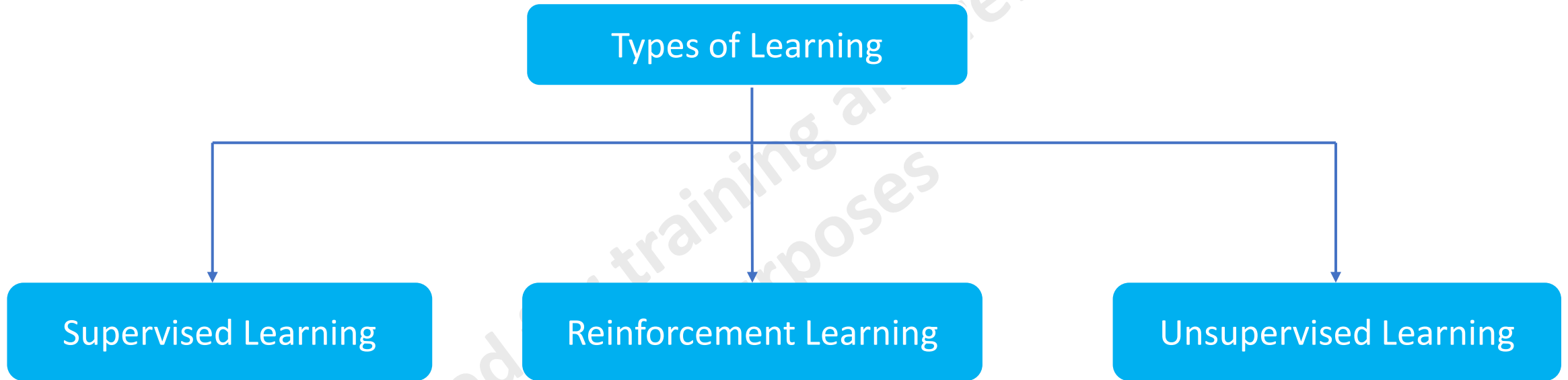
The final outcome is predicted after performing parameter tuning and improving the accuracy of the model.

Types of Machine learning

Supervised, Unsupervised ...



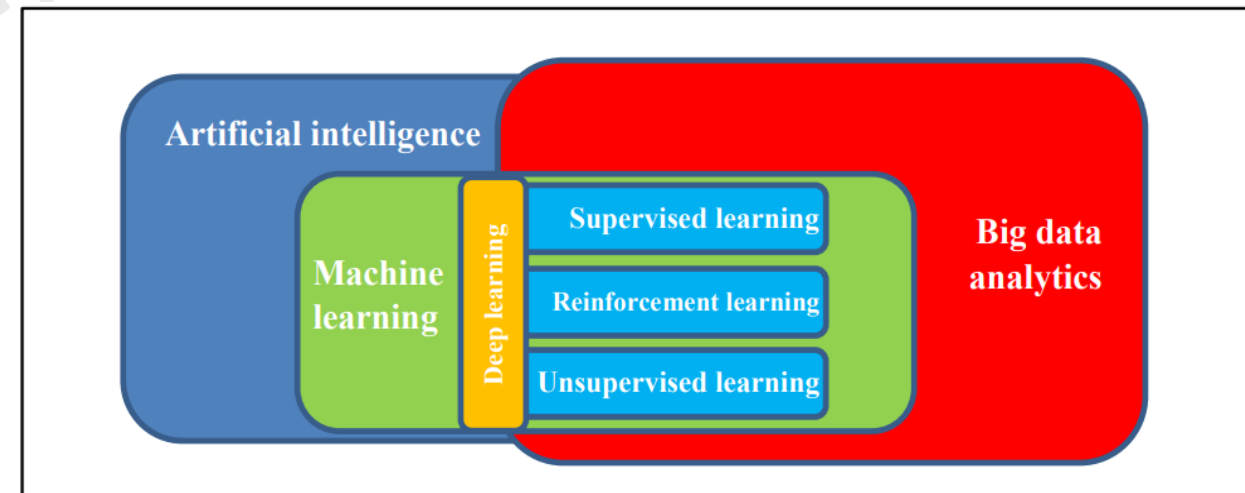
Categories of ML Algorithms



Types of machine learning

- **Supervised learning**, where observations contain input/output pairs : These sample pairs are used to "train" the ML system to recognize certain rules for correlating inputs to outputs.
- **Unsupervised learning**, where those labels are omitted: In this form of ML, rather than being "trained" with sample data, the machine learning system finds structures and patterns in the data.
- **Semi-supervised learning** –
- **Reinforcement learning**, where evaluations are given about how good or bad a certain situation
- **Deep learning**

Supervised Learning	> Labeled data > Direct feedback > Predict outcome/future
Unsupervised Learning	> No labels > No feedback > Find hidden structure in data
Reinforcement Learning	> Decision process > Reward system > Learn series of actions



SUPERVISED LEARNING

- Supervised learning is a type of machine learning where the algorithm is **trained on a labeled dataset**
- meaning the input data is paired with corresponding desired output or target values.
- The goal of supervised learning is to learn a mapping from inputs to outputs
- This creates another program called **learnt model**
- Use this outcome model to make accurate predictions or decisions on new, unseen data.



Labeled dataset

- Dataset for classifying emails as spam or not spam

Email Text	Label
-----	-----
Get a discount on your next purchase!	Spam
Meeting tomorrow at 10 AM, don't forget!	Not Spam
Win a free vacation! Claim your prize now.	Spam
Important updates on your account.	Not Spam

Labeled dataset

- Dataset for classifying iris flowers into species

Sepal Length	Sepal Width	Petal Length	Petal Width	Species
5.1	3.5	1.4	0.2	Setosa
6.4	3.2	4.5	1.5	Versicolor
7.3	2.9	6.3	1.8	Virginica

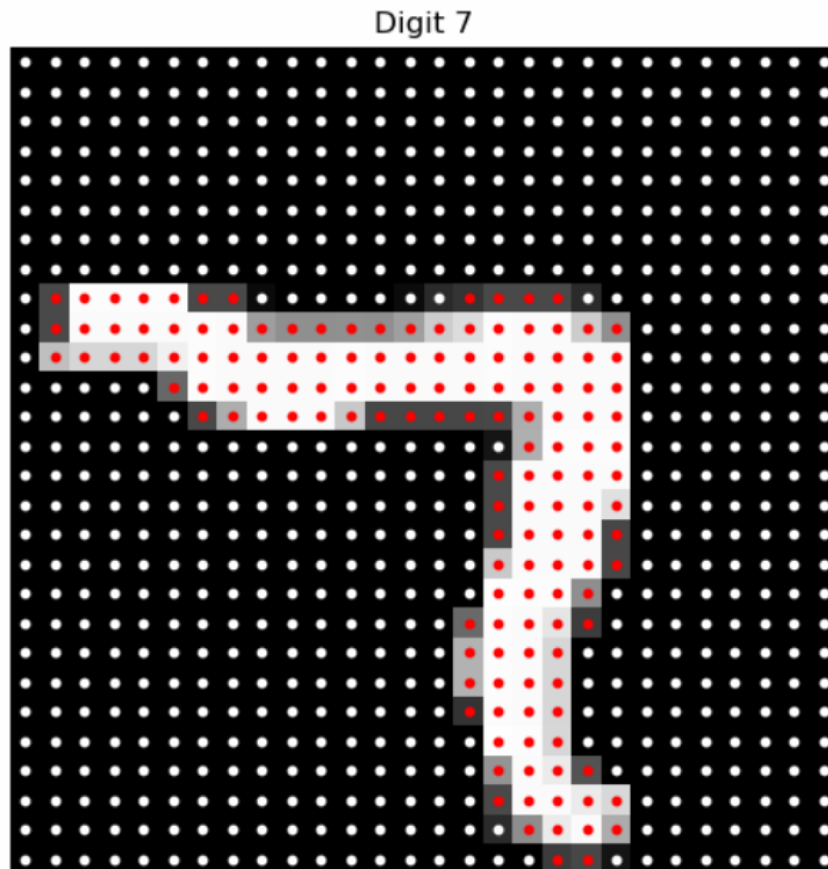
Labeled dataset

- Dataset for predicting house prices based on various features

Area (sq ft)	Bedrooms	Bathrooms	Garage Size	Price (\$)
1500	3	2	2	250,000
2000	4	3	3	350,000
1200	2	1	1	180,000

Labeled dataset

Dataset for classifying handwritten digits (0-9)



Pixel 1	Pixel 2	...	Pixel n	Digit Label
0.1	0.5	...	0.8	3
0.9	0.2	...	0.6	7
0.3	0.7	...	0.4	1

Label: 7 Label: 6 Label: 9 Label: 2 Label: 9

Labeled dataset

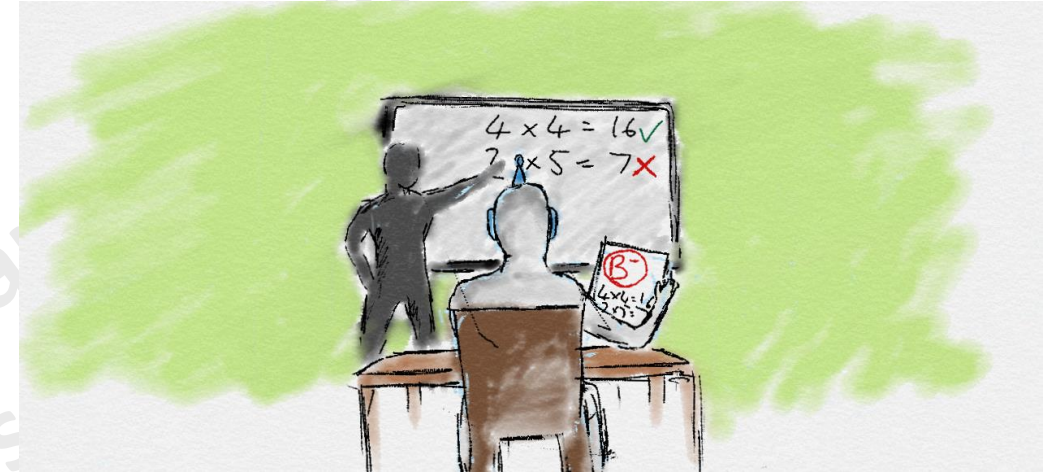
The "Text" column contains various text samples.

The "Category" column represents the corresponding category or label for each text sample.

Text	Category
Machine learning is fascinating.	Technology
The sun sets over the mountains.	Nature
Soccer is a popular sport worldwide.	Sports
The economy is experiencing a downturn.	Business
A delicious recipe for chocolate chip cookies.	Food
Shakespeare's plays are timeless classics.	Literature
....	
Hiking trails offer stunning views of the landscape.	Nature
Stock markets are influenced by global events.	Business
Pizza is one of the most loved fast foods.	Food

supervised learning

- the goal is to **learn the mapping** (the rules) between a set of inputs and outputs.
- the algorithm is fed a set of 'training' data that contains **labels** and **data**
 - For instance, a data set of transactions may contain **labels** on some data points identifying those that are fraudulent and those that are not fraudulent.
- The algorithm will 'learn' a general rule of **classification** that it will use to predict the labels for the new/unknown observations



Past labelled data	Weather/ temp (deg C)		Holiday (Y/N)	# of visitors
	20		Y	2000
	27		N	1800
	...		...	...
	39	54	N	900
New data	Weather/ temp (deg C)		Holiday (Y/N)	# of visitors
	20		Y	??
	27		N	??
	...		...	...

Inductive (Supervised) Learning

- Basic Problem: Induce a representation of a function (a systematic relationship between inputs and outputs) from examples.
- **target function** $f: X \rightarrow Y$
- **example** $(x, f(x))$
- **hypothesis** $g: X \rightarrow Y$ such that $g(x) = f(x)$
- x = set of attribute values (***attribute-value representation***)
- Y = set of discrete labels (***classification***)

Where we use supervised learning



image
classification,



speech recognition,



sentiment analysis,



recommendation
systems,



fraud detection,
and more,

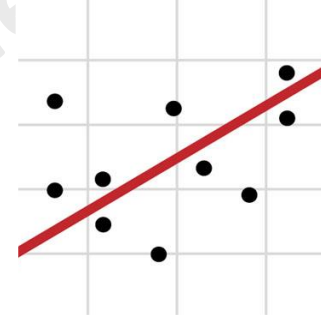
Types of supervised learning



Classification

the goal is to predict a categorical label or class for a given input based on historical data.

input data consists of features, and the algorithm's task is to assign each input to a predefined class or category.



Regression

it's used to predict a continuous numeric value based on input features.

the algorithm learns a mathematical relationship between the input features and the target value, which is a continuous number.

Example (classification)



Spam Detection

Input Features: Email content, sender information, subject line.

Categories: "Spam" or "Not Spam."

Task: Determine whether an incoming email is likely to be spam or not, helping filter out unwanted messages from inboxes.



Churn Prediction

Input Features: Customer behavior, usage patterns, engagement.

Categories: "Churn" (customer leaves) or "Not Churn."

Task: Predict whether a customer is likely to stop using a service or product, enabling proactive retention strategies.



Sentiment Analysis

Input Features: Textual data (social media posts, reviews, etc.).

Categories: "Positive," "Negative," or "Neutral."

Task: Analyze the sentiment expressed in text and determine the emotional tone or opinion conveyed.



Dog Breed Detection

Input Features: Image pixels representing a dog's photo.

Categories: Various dog breed labels.

Task: Identify the specific breed of a dog in an image, showcasing the capability of algorithms to recognize complex visual patterns.



Disease Diagnosis

Input Features: Medical test results, patient characteristics.

Categories: Different medical conditions (e.g., "Diabetes," "Healthy").

Task: Assist medical professionals in diagnosing diseases based on patient data and medical history.

Example (regression)



House Price Prediction:

Input Features: Square footage, number of bedrooms, location.

Target: Predicted price of the house.

Task: Estimate the price of a house based on its characteristics, assisting in real estate pricing and transactions.



Temperature Forecasting:

Input Features: Historical weather data, time of day, season.

Target: Predicted temperature for a specific time.

Task: Forecast the temperature for future time intervals, aiding in weather predictions and planning.



Stock Price Prediction:

Input Features: Historical stock prices, market indicators.

Target: Predicted stock price.

Task: Estimate the future value of a stock to guide investment decisions and financial strategies.



Fuel Efficiency Prediction:

Input Features: Car specifications (engine size, weight, etc.).

Target: Predicted fuel efficiency (miles per gallon).

Task: Determine the expected fuel efficiency of a vehicle, assisting consumers in making informed choices.



Health Parameter Prediction:

Input Features: Lifestyle factors, diet, exercise frequency.

Target: Predicted health parameter (e.g., blood pressure).

Task: Estimate health indicators based on individual habits, contributing to personalized healthcare advice.

ML – process flow

Labeled Data: start with a dataset where each data point consists of both the input features and the corresponding correct output (target) value.



Training: The algorithm learns from the labeled data by finding patterns and relationships between the input features and the target values.



Prediction: Once the algorithm is trained, it can take new, unseen data with input features and use the learned patterns to predict or infer the corresponding output or target value.

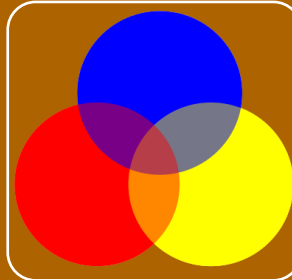


Evaluation: The performance of the algorithm is assessed using metrics like accuracy, precision, recall, or others, depending on the specific problem.

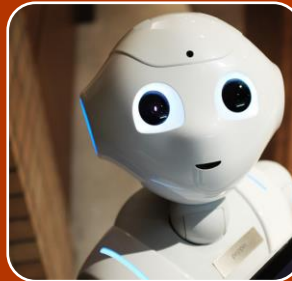
What is unsupervised learning



Imagine you have a big box of different colored marbles, but you don't know anything about them –



you don't know which colors are important or which ones belong together.



Unsupervised learning is like having a smart friend who can help you make sense of these marbles.

Examples of unsup learning



Disease Subtyping and Patient

Stratification: group patients with similar medical characteristics, genetic profiles, or disease symptoms.

help doctors understand different subtypes of diseases, tailor treatments, and predict disease outcomes more accurately.



Anomaly Detection and Fraud Prevention: unusual patterns in patient records or insurance claims.

examples

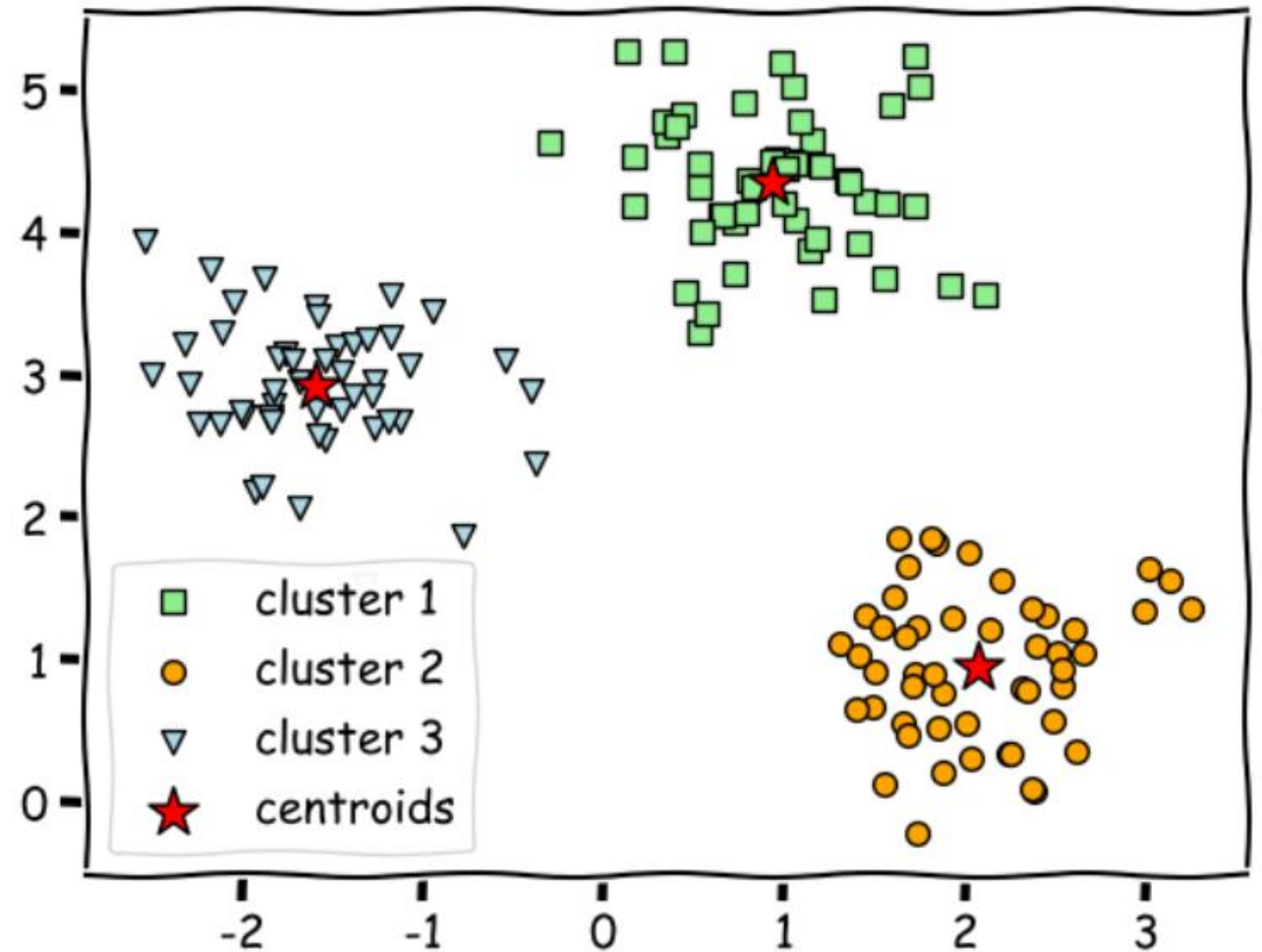


Genomic Data Analysis: With the advancement of DNA sequencing technologies, vast amounts of genomic data are generated. Unsupervised learning techniques like clustering can help group similar genetic sequences, leading to insights into genetic variations and potentially identifying genetic factors associated with certain diseases.

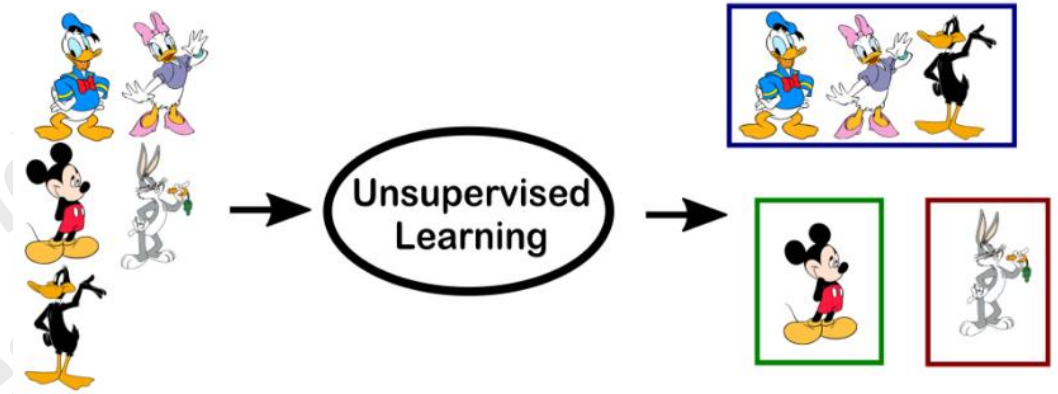
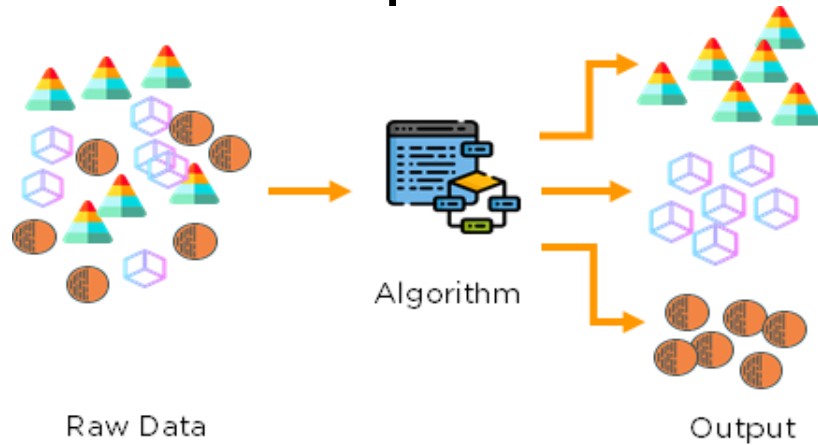


Medical Image Analysis: In radiology and pathology, unsupervised learning can be used to segment and classify regions of interest within medical images. For instance, clustering techniques can help identify regions of interest in an MRI scan or categorize cell structures in histopathology slides.

unsupervised



Unsupervised learning

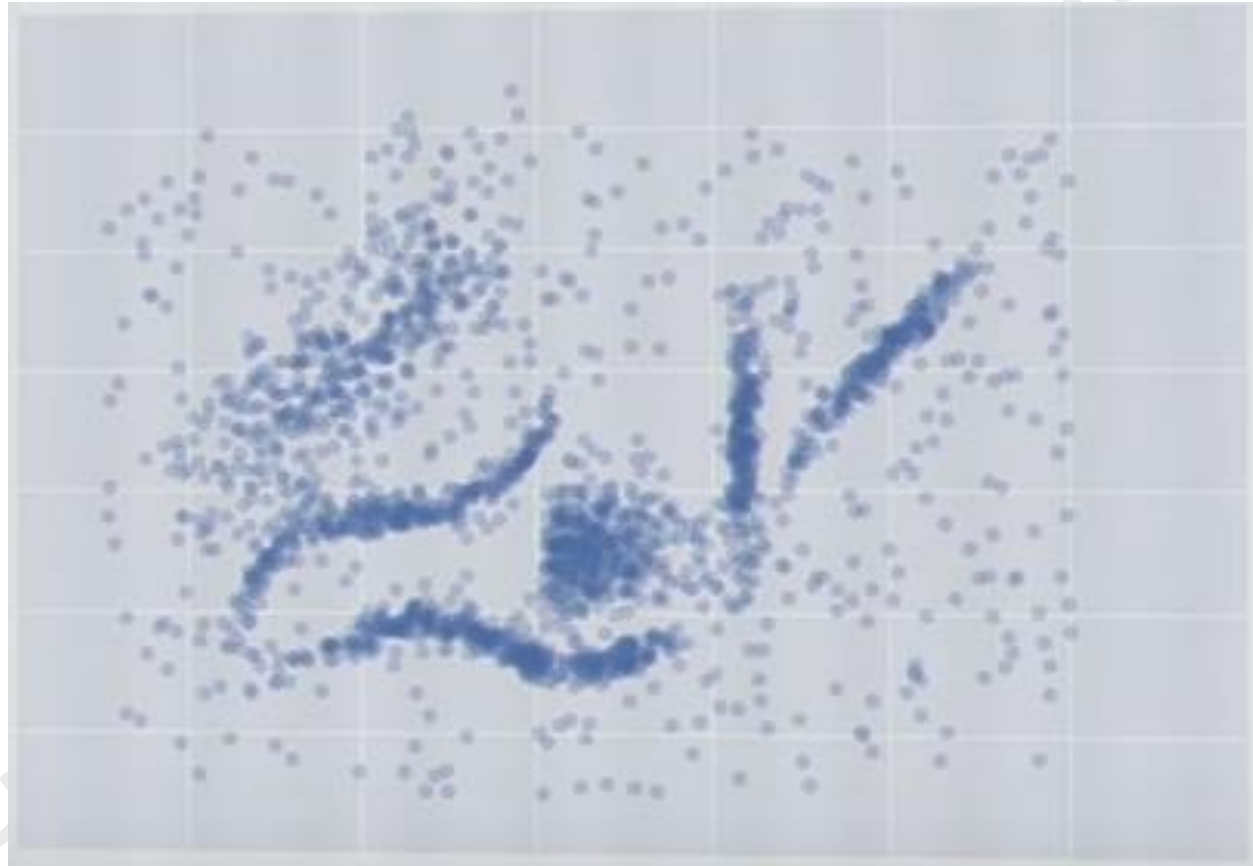


Since no labels are provided, there is **no specific way** to compare model performance in most unsupervised learning methods.

65

EASY IN THEORY BUT HARD IN PRACTICE

Human eye and grouping

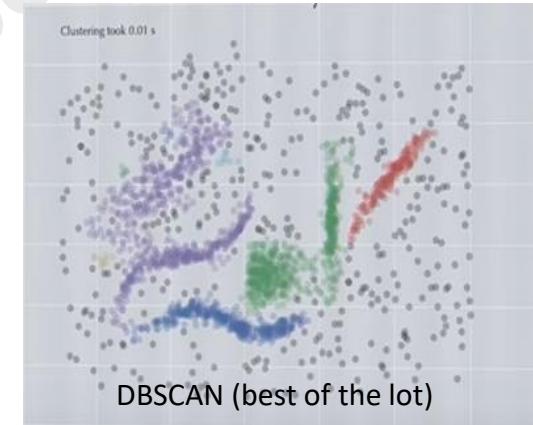
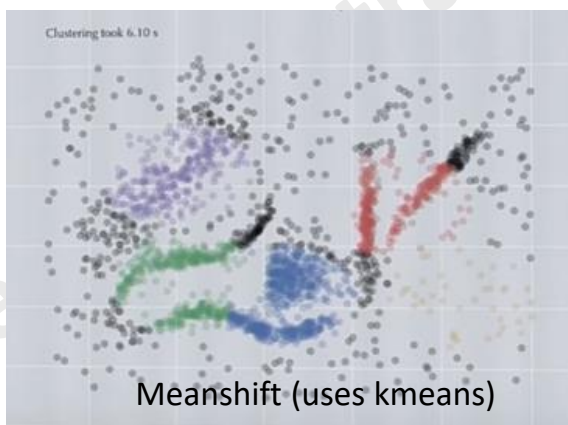
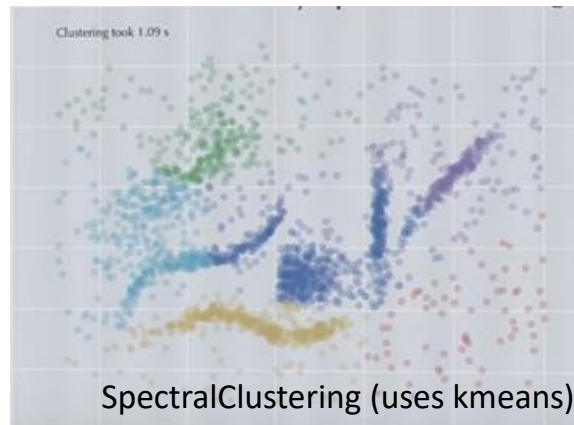
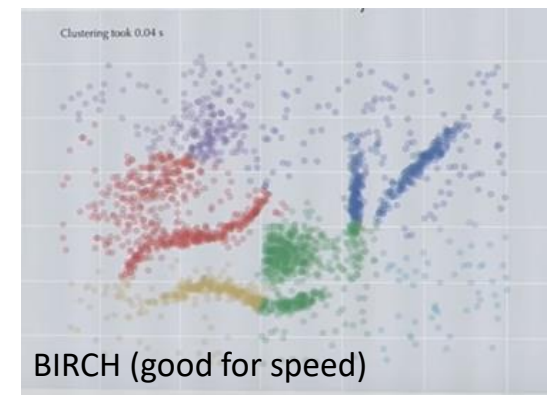
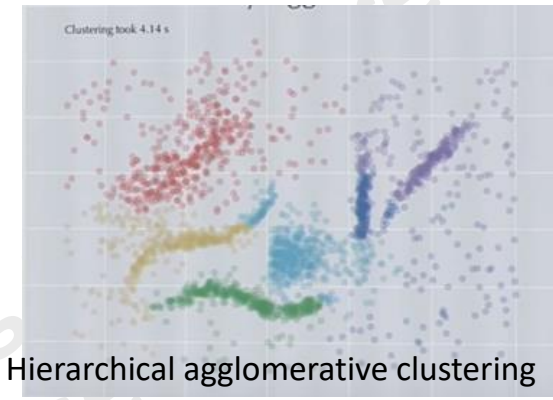
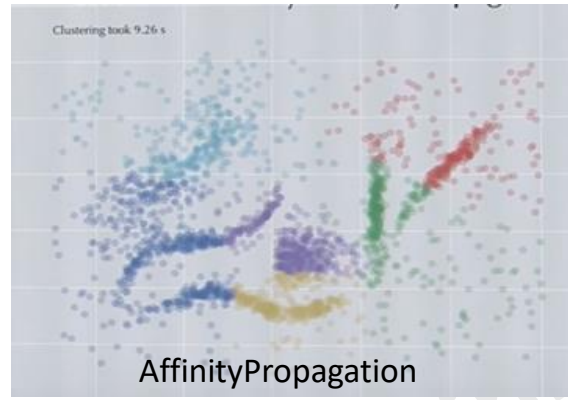
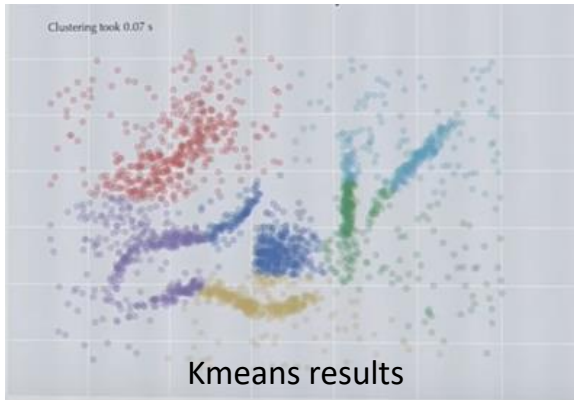


Human eye can easily pick the clusters here depicted in 2D

Easy !!!

@ Copyright – created by bhupen

Results by various clustering techniques



LET US TRY TO DEFINE REINFORCEMENT LEARNING



Reinforcement Learning (RL) is a way to **teach** computers to learn by trial and error.



It's like training a pet or playing a game.



The computer tries different **actions**, receives **feedback**, and **learns** to make better choices to get rewards.



The goal is to find the best actions for different situations to get the **most rewards**.

Responsible and Ethical

WHAT DOES THE AGENT LEARN IN REINFORCEMENT LEARNING

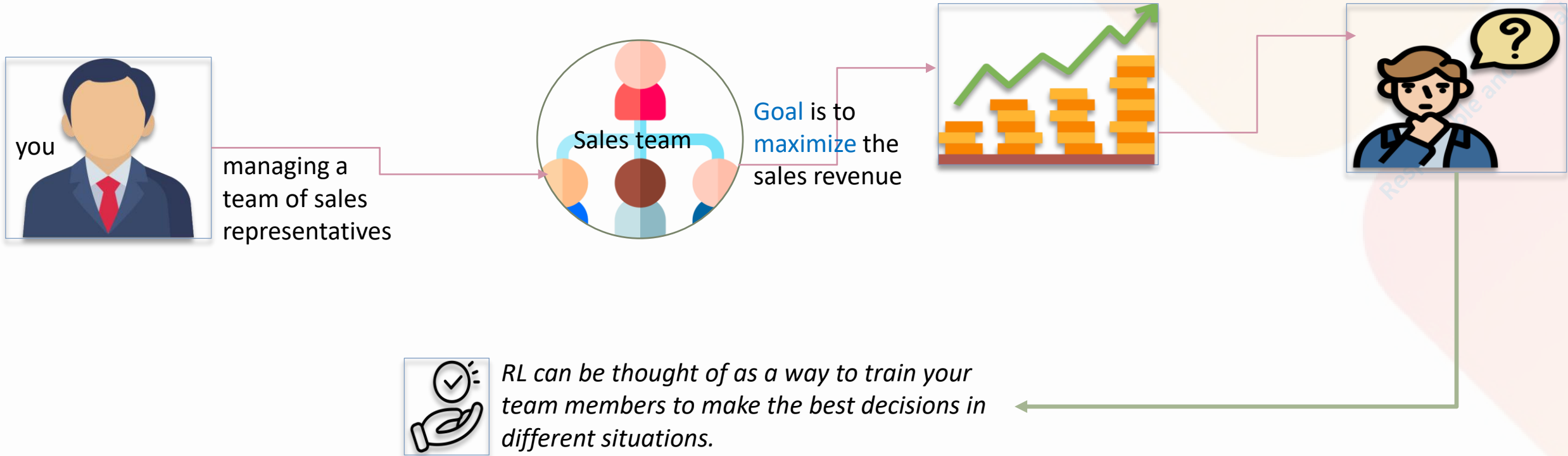
Game Playing: train agents to play games such as chess, Go, and poker.

Robotics: perform complex tasks. learn to navigate environments, manipulate objects, or even control their own movements.

Autonomous Vehicles: self-driving cars.

Recommendation Systems: RL algorithms can be used to optimize recommendations.

EXAMPLE - SALES



EXPLANATION



the team members are
the **agents**,



the **environment** is the
market or the sales
context.



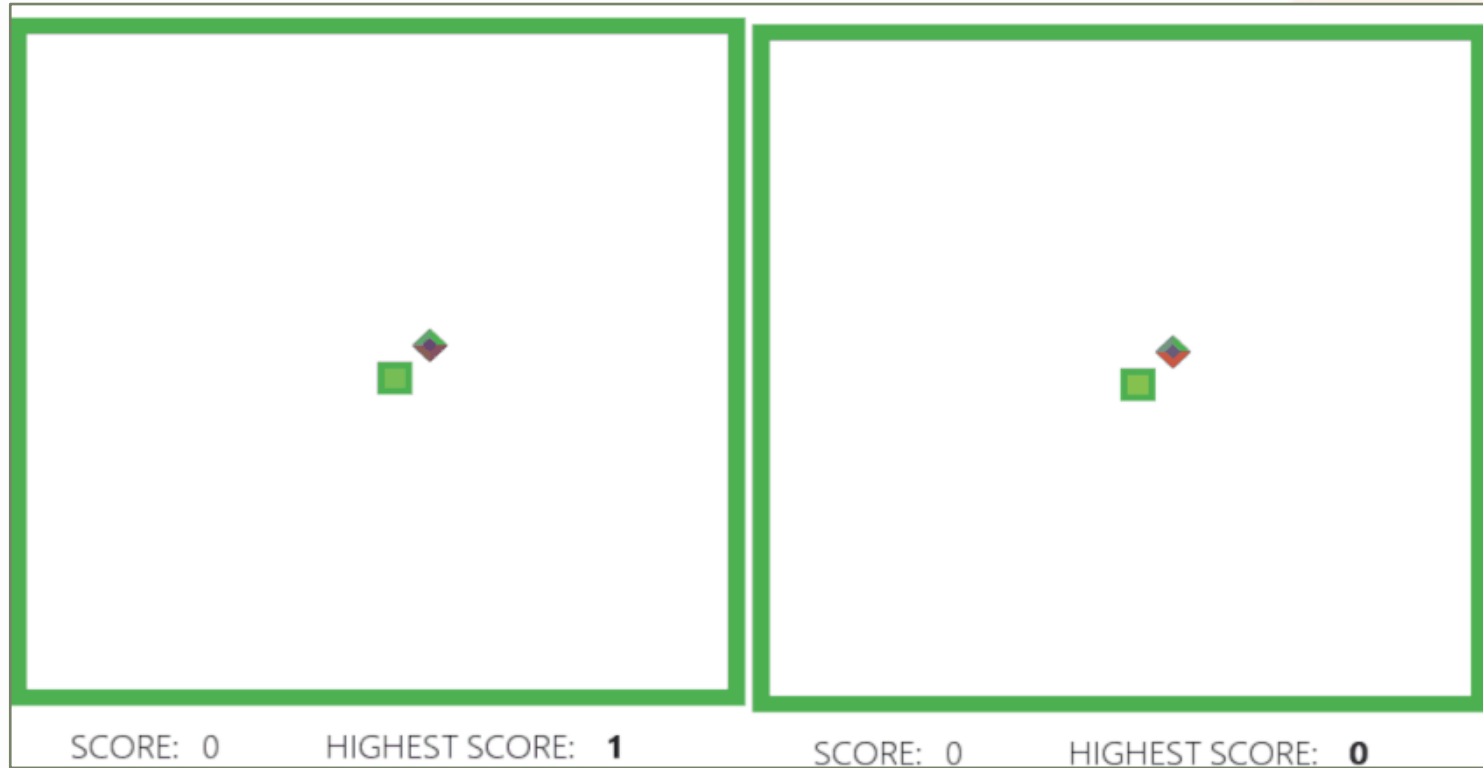
Each team member
takes **actions**

such as choosing a sales
strategy or
approaching potential
clients, and



receives feedback in the
form of **rewards** or
penalties,
immediate sales,
customer satisfaction, or
other metrics

EXAMPLE - GAMES



AI does not know anything
about the game

AI is trained and learnt how
to play.

Responsible and Ethical

HOW IS RL DEVELOPED FOR GAMES

Environment: The video game serves as the environment.

Agent: The RL agent is the player or learner.

Action Space: The set of possible actions the agent can take in the game. These actions could include moving in different directions, jumping, shooting, or performing other in-game actions.

State Space : game's current state, such as the positions of the game objects, scores, or other relevant variables.

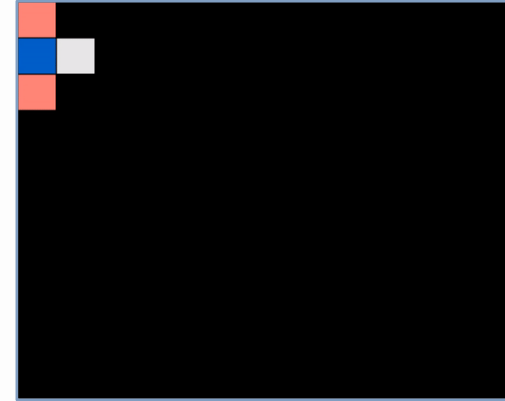
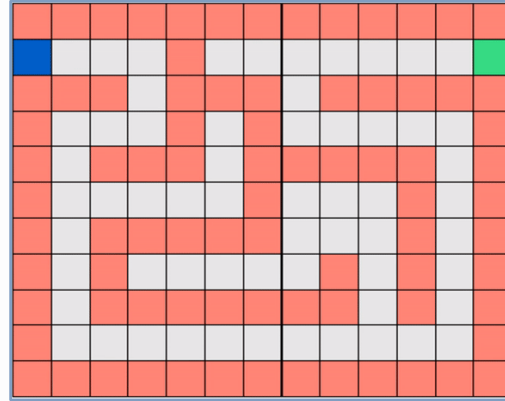
Reward Signal: The agent receives a reward signal from the environment based on its actions. In video games, rewards can be **positive** (e.g., gaining points, completing a level), **negative** (e.g., game over), or **intermediate** (e.g., collecting coins, defeating enemies). The goal is for the agent to learn to **maximize** the cumulative rewards it receives over time.

Learning Process: The RL agent learns by interacting with the game environment. It explores different actions, observes the resulting state changes, and receives rewards. The agent's goal is to discover the best actions to take in different states to maximize its long-term rewards.

EXAMPLE – GRIDWORLD

Gridworld environment

the robot can move 1 unit left, R, L, U or D at a time, to any 'safe' square (colored white).



agent cannot 'see' the entire map



Goal : robot (blue) has to reach the terminal cell (Green)



States : robot travels thru the cells, some cells robot is not allowed to go (orange cells)
Only white cells form the path



Possible decisions: the robot can go
U, D, R, L



Rewards: -1 if the robot tries to enter the Orange cells, 0 for all other steps, 1 for the Goal



Training (n_iters) : randomly takes a step. For each possible step U, D, R, L, agent will compute the numeric value (goodness) of the step in a particular cell (state)

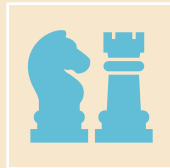


Decision making (prediction) : max of the goodness values

EXAMPLE - GAME PLAYING AI (CHESS)



a **state** : refers to a permissible arrangement of the remaining white and black pieces on the board

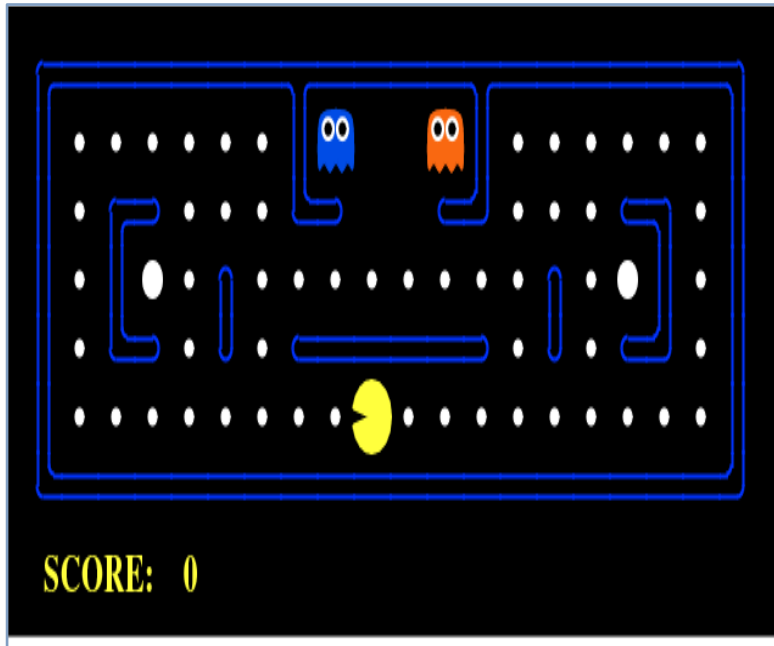


an **action** : denotes a legal move executed by any of the current pieces based on the rules of chess.

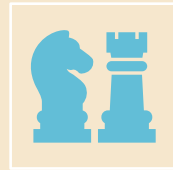


a potential **reward** structure : any move that does not promptly result in the goal state of checkmating the opponent incurs a negative reward of -1, whereas a move that successfully accomplishes a checkmate earns a substantial positive reward, such as 10,000.

EXAMPLE - GAME PLAYING AI (PAC-MAN)



a **state** : refers to a permissible arrangement player/objects/enemies on the board



an **action** : denotes a legal move executed by any of the current.



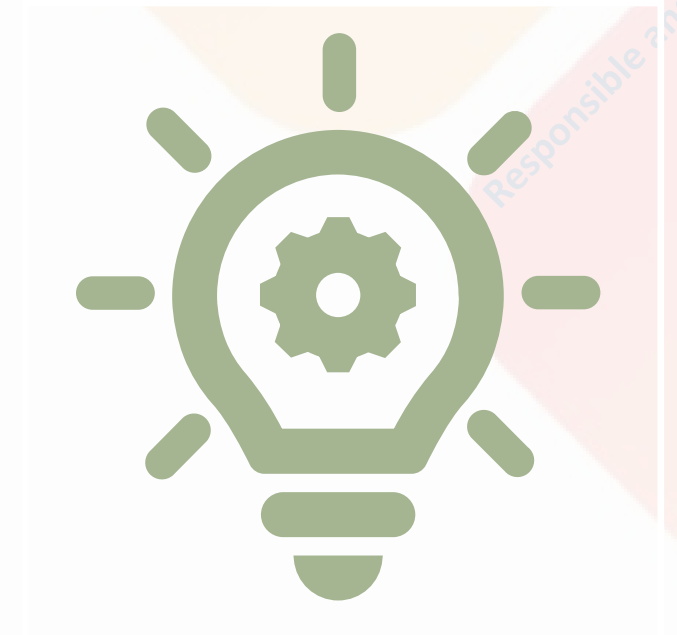
a potential **reward** structure : design

NOW ... LETS UNDERSTAND WHAT IS REINFORCEMENT LEARNING

Reinforcement learning is learning what to do

- how to map situations to actions
- so as to maximize a numerical reward signal.
- The learner is not told which actions to take, but instead must discover which actions yield the most reward by trying them.

Sutton and Barto, Reinforcement Learning: An Introduction



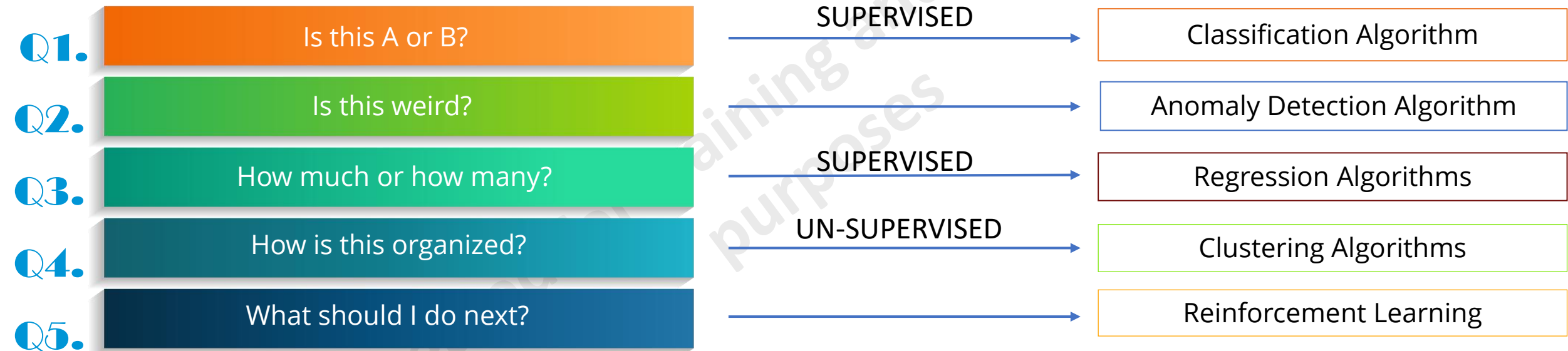
Responsible and Ethical



Typical tasks that MLs do

List is very long ... only a few are listed

Typical problems solved by machine learning



These algorithms are fitted into three types of categories, which are the following:

Is this A or B? - two-class classification

01

Will this customer
renew his/her
subscription?

02

Is this an image of a
cat or a dog?

03

Will this customer
click on the top
link?

04

Will this tyre fail in
the next thousand
miles?

05

Does the RS. 5
coupon or the 25%
off coupon result in
more return
customers?

Is this A or B or C or D? – multi-class classification



Which animal is in this image?



Which aircraft is causing this radar signature?



What is the topic of this news article?



What is the mood of this tweet?



Who is the speaker in this recording?

Is this bizarre, weird or uncommon? – anomaly detection



credit card fraud detection.



Is this pressure reading unusual?



Is this combination of purchases very different from what this customer has made in the past?



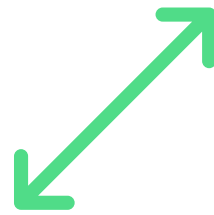
Are these voltages normal for this season and time of day?

82

How Much / How Many?



What will the temperature be next Tuesday?



What will my fourth quarter sales in Kolkata be?



How many new followers will I get next week?



Out of a thousand units, how many of this model of bearings will survive 10,000 hours of use?

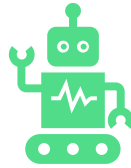
Few other applications



Recommendation Systems:

Problem: Suggesting items to users based on their preferences and behaviors.

Example: Movie recommendations on streaming platforms, product recommendations on e-commerce websites.



Natural Language Processing (NLP):

Problem: Processing and understanding human language.

Example: Language translation, chatbots, sentiment analysis, text summarization.



Computer Vision:

Problem: Analyzing and extracting information from visual data.

Example: Object detection, facial recognition, autonomous vehicles.

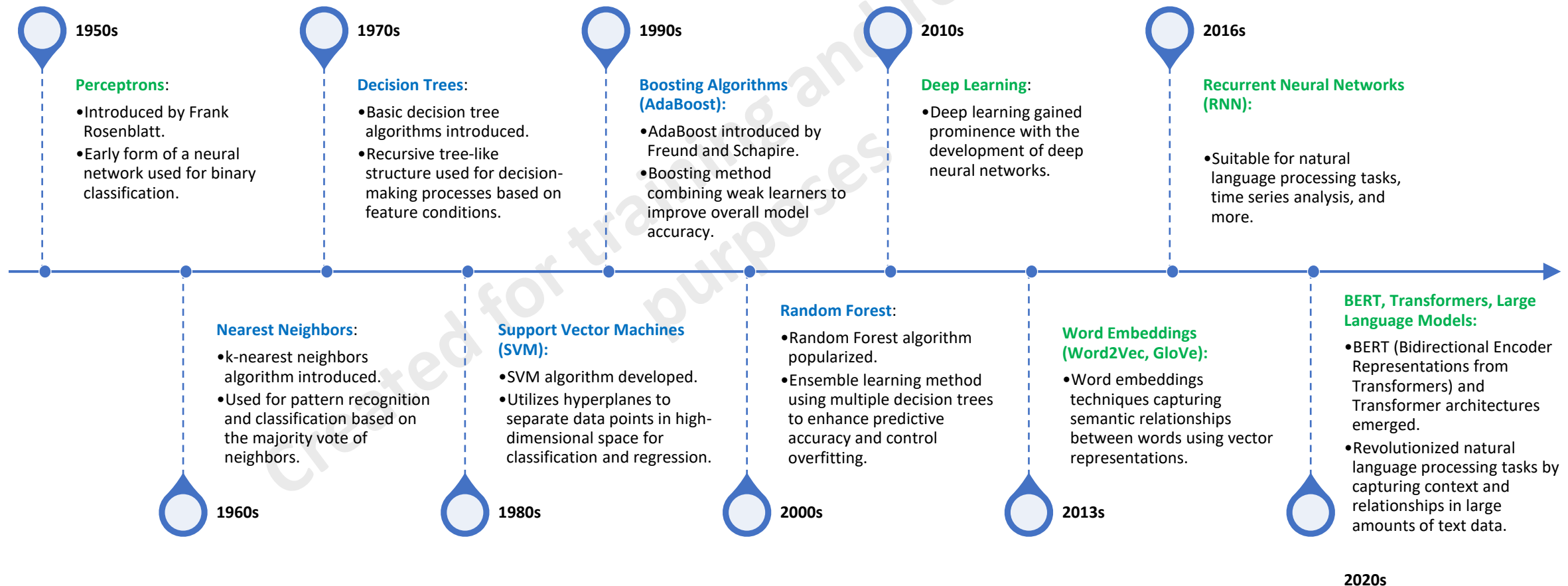


Time Series Analysis:

Problem: Predicting values over time based on historical data.

Example: Stock price prediction, weather forecasting, energy consumption prediction.

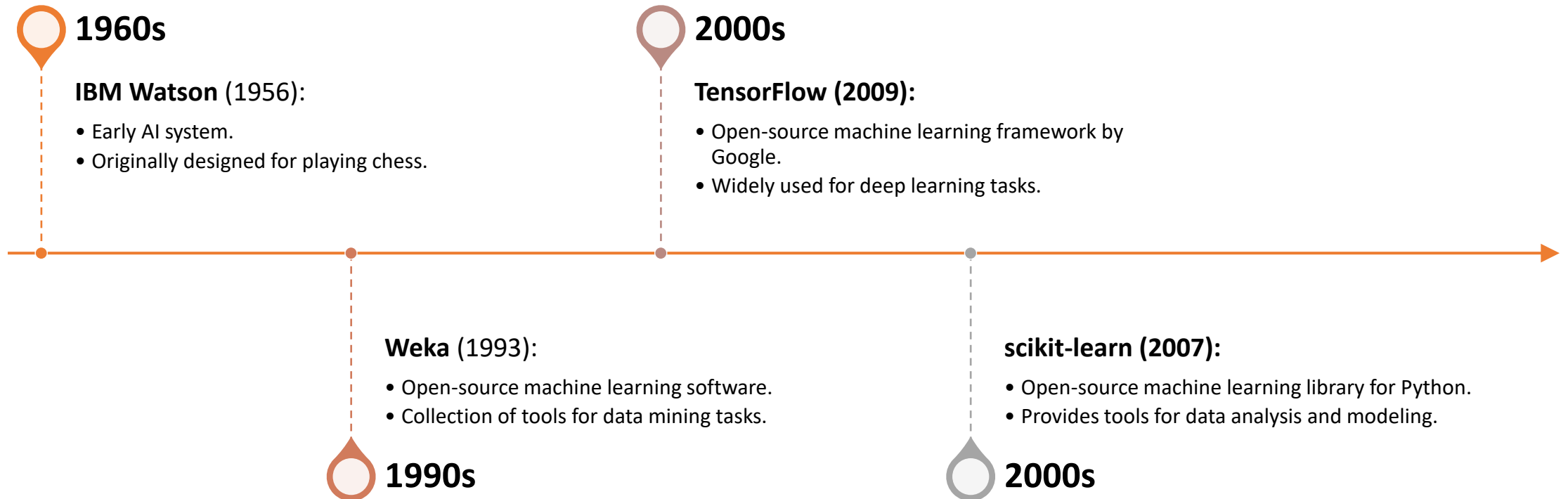
Evolution of ML algorithms (just a glance)



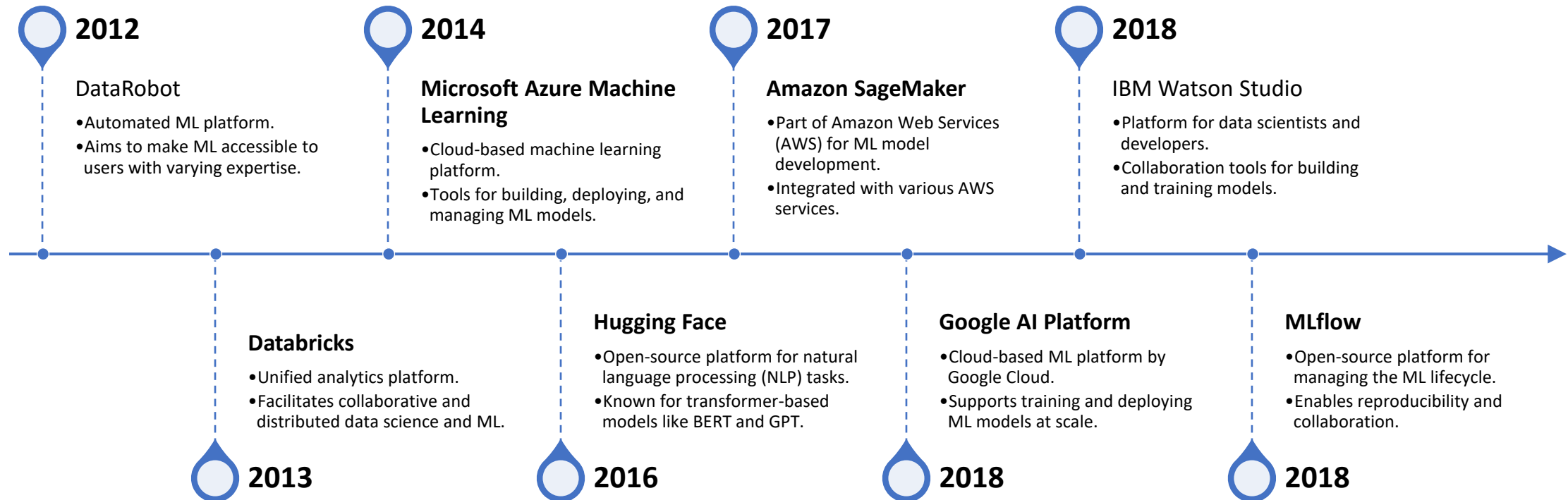
Popular ML platforms

A thick, hand-drawn style orange line that underlines the title 'Popular ML platforms'.

Popular ML platforms



Popular cloud-based ML platforms



programming languages used in ML

Python:

- **Strengths:** Python is the most widely used language in machine learning. It has extensive libraries and frameworks (e.g., TensorFlow, PyTorch, scikit-learn) and is known for its readability and ease of use.
- **Applications:** General-purpose machine learning, deep learning, data analysis.

R:

- **Strengths:** R is a statistical programming language with rich data visualization capabilities.
- **Applications:** Statistical analysis, data visualization, predictive modeling.

Java:

- **Strengths:** also used for building machine learning models.
- **Applications:** Enterprise-level applications, machine learning in Java.

Some more ...

C++:

- **Strengths:** C++ is a high-performance language and is used when efficiency is crucial, especially in applications like computer vision and game development.
- **Applications:** Computer vision, robotics, high-performance computing.

JavaScript (with TensorFlow.js):

- **Strengths:** JavaScript is widely used for web development. TensorFlow.js allows running machine learning models directly in the browser.
- **Applications:** Web-based machine learning applications.

Julia:

- **Strengths:** Julia is designed for high-performance numerical and scientific computing. It aims to be as fast as low-level languages while maintaining high-level abstractions.
- **Applications:** Scientific computing, data science.

Scala:

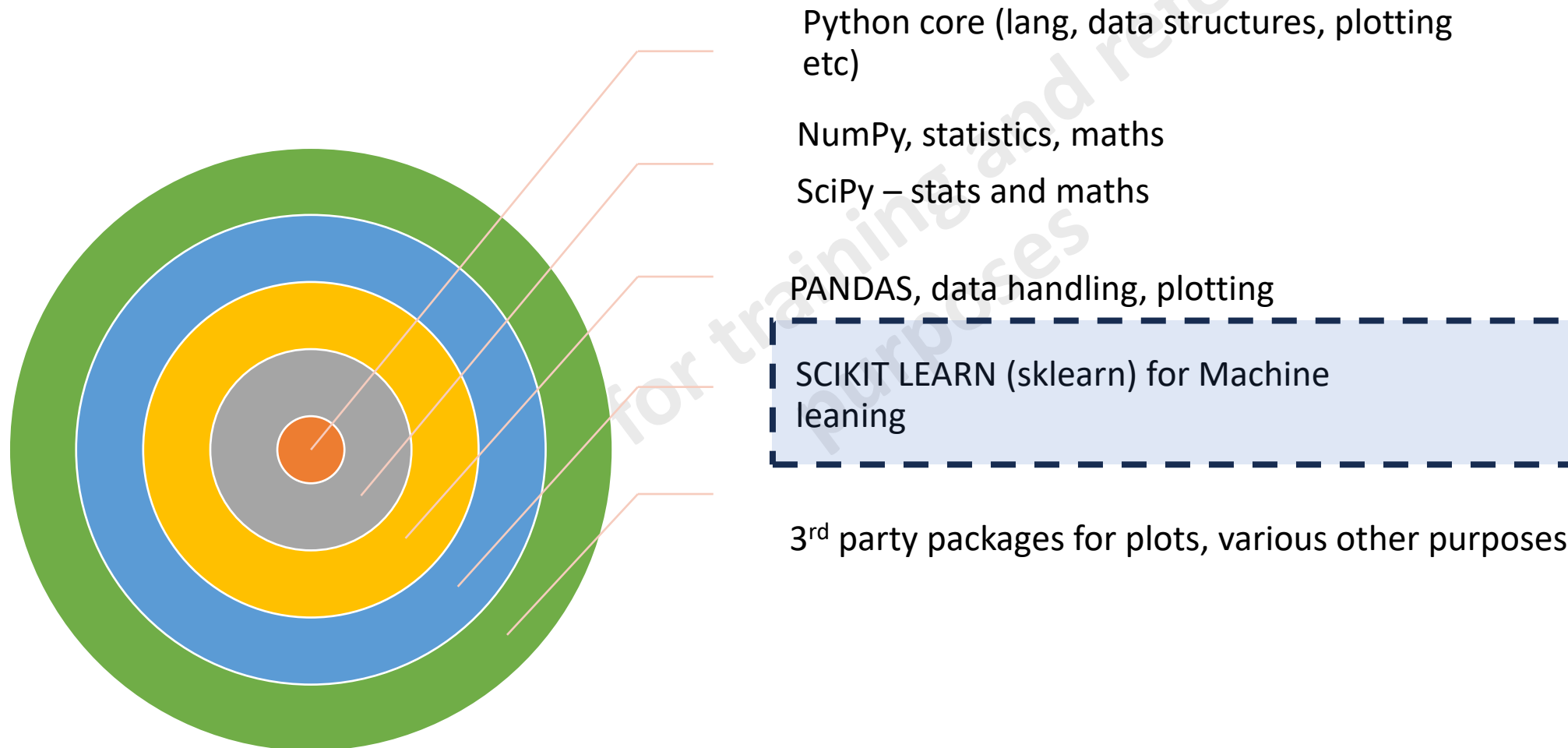
- **Strengths:** Scala is a general-purpose programming language that runs on the Java Virtual Machine (JVM). It is used for scalable and distributed machine learning applications.
- **Applications:** Distributed computing, scalable machine learning.



Python components

For Machine learning

Components stack for classical MLs (Python)



How do we use SCIKIT learn module?

Understand the Problem

- Objective, scope, users, security, performance, scale etc

Data Exploration

- characteristics of your data, including the types of features, the distribution of classes (for classification), and potential relationships.

Data Preprocessing

- such as handling missing values, scaling features, and encoding categorical variables.

Choose a Model

- Select a machine learning model based on your problem. (BASIS FOR SELECTING A MODEL)

Model Training

- Train the chosen model using the training data. The model learns the patterns and relationships within the data.

Make Predictions

- Use the trained model to make predictions on the test data. This step helps evaluate how well the model generalizes to new, unseen examples.

Model Evaluation

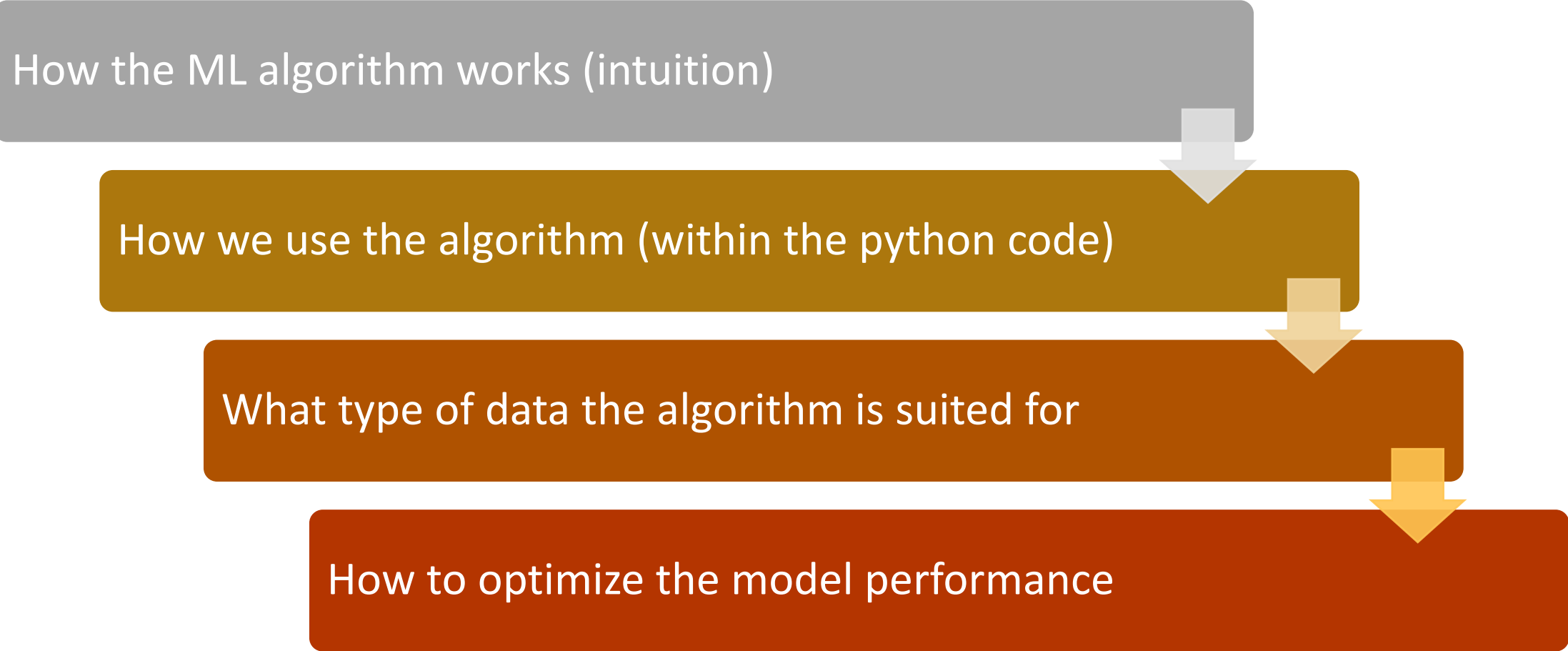
- Assess the model's performance using metrics such as accuracy, precision, recall, and F1-score for classification tasks, or Mean Squared Error for regression tasks.

Fine-Tuning (Optional)

- If needed, fine-tune the model by adjusting hyperparameters or using techniques like cross-validation.

For choosing a model

How the ML algorithm works (intuition)



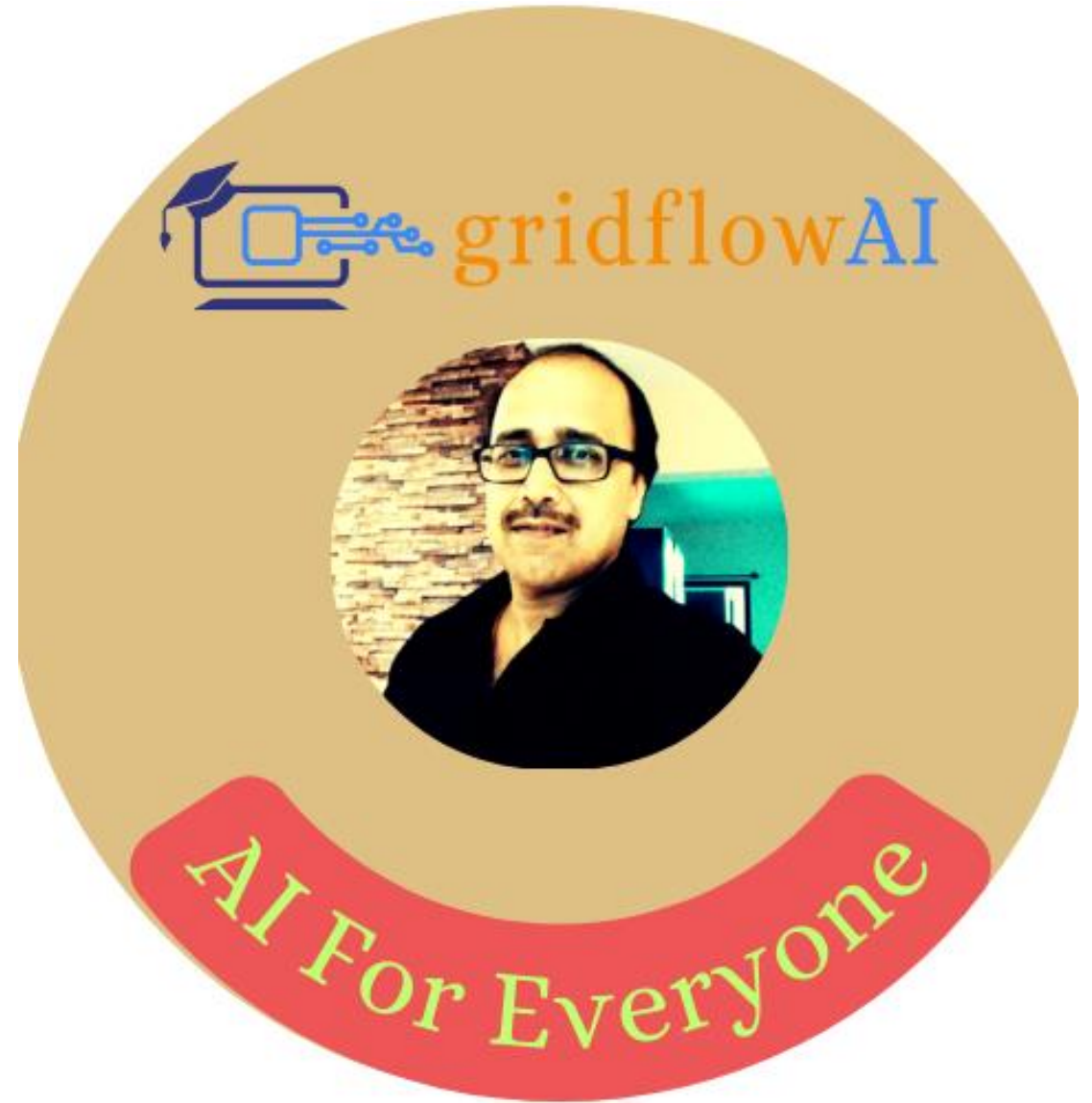
```
graph TD; A[How the ML algorithm works (intuition)] --> B[How we use the algorithm (within the python code)]; B --> C[What type of data the algorithm is suited for]; C --> D[How to optimize the model performance];
```

How we use the algorithm (within the python code)

What type of data the algorithm is suited for

How to optimize the model performance

Thanks &
Happy
Learning !!



Demo using python/sklearn (Ridge and Lasso)

